Pablo Piantanida

Academic Curriculum Vitae

Montreal, QC, Canada
Paris-Saclay, France
E *pablo.piantanida@mila.quebec*
W *pablo-piantanida.org*

Executive Profile

Full Professor at CentraleSupélec, Université Paris-Saclay, and senior researcher developing the **mathematical and information-theoretic foundations of reliable, adaptive, and collaborative intelligence.** My research studies what information is relevant to a task, how it should be represented, acquired, and exchanged, what can be inferred from limited observations, and how intelligent systems should reason and act under uncertainty and resource constraints. It connects information theory, statistical decision theory, statistics, and modern machine learning to develop fundamental results, actionable methods, and experimentally validated approaches to learning, reasoning, evaluation, and decision-making. Since 2022, I have served as **Director of the CNRS International Research Laboratory on Learning Systems** in Montréal, where I shape the scientific strategy, develop institutional partnerships, and coordinate France–Canada research initiatives connecting foundational AI with language, healthcare, scientific discovery, and engineering.

My research has evolved from the fundamental limits of communication and distributed inference toward information for learning, reasoning, and reliable decision-making. My contributions span task-relevant representations, information estimation and privacy, uncertainty and failure prediction, generative AI evaluation, and pragmatic multi-agent communication. A common objective is to determine what information is sufficient for a task and when additional information, computation, or interaction has decision value.

My 2027–2031 programme develops three interconnected directions: **learning, representation, and reasoning**; **uncertainty, evaluation, and reliability**; and **collaborative and agentic intelligence**. As AI systems increasingly reason, retrieve information, use tools, communicate, and act in complex environments, the challenge is to determine **what information to acquire, what computation to perform, what to communicate, and when to act or defer**. The programme seeks fundamental limits, decision-theoretic principles, and resource-aware algorithms, with validation in language, scientific discovery, healthcare, and engineering.

Research Impact at a Glance

Publications	**253 publications**: 60 journal articles (predominantly Q1) and 193 international conference papers, including numerous papers in flagship (A*) conferences.
Research funding	EUR 2.31M + CAD 504K + USD 60K in competitive funding secured as PI, co-PI, co-holder, or chair holder, plus EUR 120K in in-kind GPU resources.
Industry	Recent research collaborations with **GE HealthCare, Mistral AI, Safran, IBM, Ekimetrics, and Air Liquide** on medical imaging, AI-assisted control, tabular foundation models, and generative AI for molecular generation.
Doctoral supervision	**22 completed Ph.D. students** and **13 current Ph.D. students**, supervised or co-supervised.
Profiles	Google Scholar | DBLP | LinkedIn | ORCID: 0000-0002-8717-2117
Impact	h-index: 36; 5,900+ citations (Google Scholar).

CV Contents

Current and Previous Appointments . . . 2
Research and Institutional Leadership . . . 2
Competitive Funding and Research Support . . . 3
Doctoral Supervision and Research Mentoring . . . 4
Selected Honors and Distinctions . . . 8
Publication Record . . . 9
International Scientific Leadership and Service . . . 9
Teaching, Graduate Training and Curriculum Leadership . . . 11
Education and Degrees . . . 12
Invited Seminars, Tutorials and Outreach . . . 12
Full Publication List . . . 14

Current and Previous Appointments

January 2022 – Present **Director, CNRS International Research Laboratory on Learning Systems (ILLS, IRL2020)**, *International Research Laboratory jointly supported by McGill University, ÉTS, Mila, Université Paris-Saclay, and CentraleSupélec*, Montréal, QC

Sep. 2023 – Present **Associate Academic Member**, *Mila - Quebec AI Institute*, Montréal, QC, Canada

Sep. 2026 – Present **Adjunct Professor**, *McGill University*, Montréal, QC, Canada

Sep. 2022 – Present **Adjunct Professor**, *École de technologie supérieure (ÉTS)*, Montréal, QC, Canada

Sep. 2020 – Present **Full Professor**, *CentraleSupélec - Université Paris-Saclay*, France

Sep. 2019 – 2022 **Associate Research Memeber**, *Comète Inria Team (Laboratoire d'informatique de l'École polytechnique)*, École polytechnique, Palaiseau, France

Sep. 2019 – 2020 **Visiting Researcher**, *Laboratoire de Mathématiques d'Orsay (LMO)*, Orsay, France

June 2018 – 2019 **Visiting Researcher**, *Université de Montréal (UdeM), Mila - Quebec AI Institute*, Montréal, QC, Canada

Sep. 2015 – 2020 **Associate Professor**, *École CentraleSupélec, Laboratoire des Signaux et Systèmes (L2S, UMR 8506), CNRS, Université Paris Saclay, France*

Sep. 2007 – 2015 **Assistant Professor**, *École Supélec, Département de Télécommunications, France*

Research and Institutional Leadership

2022 – Present **Director and founding scientific lead**, *CNRS International Research Laboratory on Learning Systems (ILLS, IRL2020)*, Montréal, Canada, Joint laboratory involving McGill University, ÉTS, Mila, CNRS, Université Paris-Saclay, and CentraleSupélec
Lead the laboratory's international research strategy, institutional partnerships, scientific programme, and France–Canada research and training initiatives

2017 – 2020 **Scientific Coordinator of the IRP (International Research Project) "Information, Learning and Control"**, *France–Canada collaborative research network*, 8 Canadian and French institutions: McGill University, Université de Montréal, ÉTS, Concordia University, INRS, Université Paris-Sud, CNRS, and CentraleSupélec, Coordination of an international research network in information, learning, and control, laying the groundwork for long-term transatlantic collaboration

2017 – 2018 **Curriculum Coordinator, ST4 track, SCOC program**, *CentraleSupélec*, Led first-year course and curriculum design within the engineering program

2018 – 2019 **Co-coordinator of the International Summer School on Artificial Intelligence**, *Direction Internationale, CentraleSupélec*

2015 – 2019 **Coordinator, research group ITA**, *Information Theory and its Applications (7 staff members out of 15)*, Laboratoire des Signaux et Systèmes, CNRS-UMR8506

2015 – 2019 **Co-responsible for CentraleSupélec of Master**, *Multimedia Networking*, E3A / Electrical Engineering, Université Paris Saclay

Competitive Funding and Research Support (selected)

2025 – 2029 **ANR–Canada Tri-agency joint call on AI (ANR; NSERC/SSHRC/CIHR) with IVADO support**, *PI*, "REGARD — Robust Evaluation of Generative AI in Real-World Environments"
Collaboration between University of Waterloo, McGill University, Aix-Marseille Université, CNRS, CentraleSupélec, and Orange Labs.

2026 - 2030 **Holder of the AI Cluster DATAIA Paris-Saclay - Chaires modulaires**, *Mathematical Principles of Learning and Communication in Attention-Based Models*, Jointly with Prof. Claire Boyer and Etienne Boursier at Laboratoire des Mathématiques d'Orsay (LMO, UPSaclay)

2025 – 2027 **IVADO R2 Group**, *co-PI*, "Cognitive Communications for Multi-Agent Reinforcement Learning"
Collaboration between McGill University, ETS, Université Laval

2026 – 2029 **CIFRE - GE HealthCare (industry collaboration)**, *PI*, "Foundation Models for 3D Image Segmentation"
(excluding salary and other expenses).

2026 – 2029 **CIFRE - Ekimetrics (industry collaboration)**, *PI*, "Foundation Models for Tabular Data"
(excluding salary and other expenses).

2024 – 2027 **CIFRE - Air Liquide (industry collaboration)**, *PI*, "Generative AI for Industrial Applications"
(excluding salary and other expenses).

2024 – 2026 **FRQS - HEALTH DATA HUB**, *Co-PI*, "Management of Patients with Breast Cancer"

2023 **Amazon Research Awards**, *The awards program offers unrestricted funds and AWS Promotional Credits to support research at academic institutions and non-profit organizations in areas of AI*, (website)

2022 – 2023 **Bourse de recherche Mitacs Globalink - Centre National de la Recherche Scientifique (CNRS)**, *PI*, "Making AI Ready for Safety-Critical Applications"

2021 – 2022 **Collaborative industrial project - 360Learning - (industry collaboration)**, *Co-PI*, "Collaborative Learning Score", s

2020 – 2023 **PSPC Aida (BPI France) - Université Paris Saclay - IBM France -**, *Co-PI*, Safety AI

2019 – 2022 **Grant DATAIA - Institut de convergence**, *Co-PI*, Leakage of Sensitive Training Data from Deep Neural Networks

2018 – 2020 **H2020-MSCA-IF-2017 - Marie Sklodowska-Curie Global Fellowships - European Commission**, *PI*, Information Theory beyond Communications: Distributed Representations and Deep Learning

Doctoral Supervision and Research Mentoring

My mentoring activity spans doctoral supervision, co-tutelle arrangements, postdoctoral training, and research guidance across France, Canada, US, Germany, Austria, Argentina, Chile, and international collaborations. It reflects a sustained commitment to graduate education, interdisciplinary AI research, several industry collaborations, and the development of early-career researchers.

13 Current Ph.D. students (3 graduating in 2026):

1. **Mr. Judeson Fernando Anthony**, *Ph.D. student at Technische Hochschule Ingolstadt (THI), affiliated with Kunstliche Intelligenz/Informatik*, topic: "Uncertainty Estimation in Vision-Based Perception Systems for Autonomous Driving", Started in Automne 2026
   Co-advised at 50% with Prof. Torsten Schön (THI), Germany.

2. **Mr. Guillermo Martin**, *Ph.D.* ***(co-tutelle)*** *student at Ecole de Technologie Supérieure (ÉTS) and Université Paris-Saclay, affiliated with LMF, ILLS, and Mila - Quebec AI Institute*, topic: "Belief Shaping and Strategic Communication in Multi-Agent AI Systems", Started in Winter 2026
   Co-advised at 33% with Prof. Christian Desrosiers (ILLS, ÉTS) and Prof. Benedikt Bollig (LMF, Université Paris-Saclay).

3. **Mr. Thomas Hennequin**, *Ph.D. student at Ecole de Technologie Supérieure (ÉTS), collaboration with Safran Canada*, topic: "AI-based Control Method for Aircraft Flight Assistance", Started in Autumn 2026
   Co-advised at 50% with Philippe Feyel (Safran).

4. **Ms. Cléa Chataigner**, *Ph.D* ***(co-tutelle)*** *student at McGill University and Université Paris-Saclay, affiliated with Mila - Quebec AI Institute and ILLS*, topic: "Human-In-The-Loop Auditing for Generative AI Systems", Started in Fall 2025
   Co-advised at 50% with Prof. Golnoosh Farnadi (Mila, McGill University | Google).

5. **Mr. Antoine Andurao**, *Ph.D. student at Université Paris-Saclay, CIFRE with GE Healthcare, affiliated with MICS and ILLS*, topic: "Foundation Models for 3D Medical Imaging", Started in Winter 2025
   Co-advised at 50% with Prof. Maria Vakalopoulou (MICS, CentraleSupélec).

6. **Mr. Matteo Sammut**, *Ph.D.* ***(co-tutelle)*** *student at Ecole de Technologie Supérieure (ÉTS) and Paris Dauphine University, affiliated with LAMSADE, ILLS, and Mila - Quebec AI Institute*, topic: "Toward a Theoretical Framework for Auditing Generative AI Models", Started in Fall 2025
   Co-advised at 33% with Prof. Yan Chevaleyre (LAMSADE, Paris-Dauphine University) and Prof. Rafael Pinot (LPSM, Sorbonne Université).

7. **Ms. Chiara Roverato**, *Ph.D. student at McGill University, affiliated with Mila - Quebec AI Institute and ILLS*, topic: "Diffusion Models for Tabular Data", Started in Winter 2025
   Co-advised at 50% with Prof. Mark Coates (ILLS, Mila, McGill University).

8. **Mr. Andrew McDonald**, *Ph.D. student at Université Paris-Saclay, CIFRE with Ekimetrics, affiliated with ILLS*, topic: "Building Foundation Models for Tabular Data", Started in Fall 2025
   Co-advised at 50% with Dr. Spyros Angelopoulos (ILLS, CNRS).

9. **Mr. Lautaro Estienne**, *Ph.D. student at Université Paris-Saclay, affiliated with LISN and ILLS*, topic: "Smart Conversations: Transforming Human-Machine Interaction", Started in Winter 2025
   Co-advised at 50% with Prof. Nona Naderi (LISN, Université Paris-Saclay)

10. **Ms. Zahra Dehghani Tafti**, *Ph.D. student at Ecole de Technologie Supérieure (ÉTS), affiliated with ILLS*, topic: "Machine Unlearning", Started in Autumn 2023
    Co-advised at 50% with Prof. Mohammad Hadi Shateri (LIVIA, ÉTS Montréal).

11. **Mr. Arthur Satouf**, *Ph.D. student at Paris-Saclay University, CIFRE with Air Liquide*, Topic: "Information Retrieval", Started in 2024
    Principal advisor and co-advising at 33% with Dr. Habiboulaye Amadou-Boubacar (Air Liquide) and Benjamin Piwowarski (CNRS, ISIR).

12. **Mr. Matthieu Dubois**, *Ph.D. student at Sorbonne Université, affiliated with ISIR*, topic: "Detecting AI Generated Text", Started in Fall 2023
    Co-advised at 50% with Dr. François Yvon (ISIR, CNRS).

13. **Mr. Philippe Formont**, *Ph.D* ***(co-tutelle)*** *student at Université Paris-Saclay and Ecole de Technologie Supérieure (ETS)*, Topic: "Computational Tools for Molecule Discovery", Started in Spring 2023
    Principal advisor and co-advising at 50% with Prof. Ismail Ben Ayed (ETS, LIVIA - ILLS).

22 Former Ph.D. students:

1. **Dr. Eric Aubinais (Fellow researcher at KTH)**, *Ph.D from Paris-Saclay University*, Topic: "Conformal Learning with Applications to Machine Learning", Graduated in 2025
   Co-advising at 50% with Prof. Elisabeth Gassiat (LMO, Université Paris-Saclay)

2. **Dr. Maxime Darrin (Mistral AI, France)**, *Ph.D (co-tutelle) from Paris-Saclay University and McGill University*, Topic: "Metrics and Methods for Generative AI Model Evaluation", Graduated in 2025
Principal advisor and co-advising at 50% with Prof. Jackie Cheung (McGill University, Canada).

3. **Dr. Eduardo DADALTO CAMARA GOMES (Helsing France)**, *Ph.D from Paris-Saclay University*, Topic: "Securing the use of AI", Graduated in 2024
Principal advisor with co-advising at 70% with Dr. Florence Alberge (L2S, Université Paris-Saclay).

4. **Dr. Ganesh Del Grosso (Researcher/Developer in AI and Cybersecurity at Situation.sh, France)**, *Ph.D from École Polytechnique*, Topic: "Leakage of Sensitive Training Data from Deep Neural Networks", Graduated in 2023
Co-advising at 50% with Dr. Catuscia Palamidessi (LIX, Inria).

5. **Dr. Sebastian Espinosa (Engineering Researcher, Chile)**, *Ph.D from Universidad de Chile*, Topic: "Learning Invariances in Deep Learning", Graduated in 2023
Co-advising at 50% with Prof. Jorge Silva (University of Chile, Chile).

6. **Dr. Marine Picot (Research Scientist at Digeiz, France)**, *Ph.D (co-tutelle) from Université Paris-Saclay and McGill University*, Topic: "Artificial Intelligence in Smart Grids", Graduated in 2023 from Paris-Saclay University and McGill University
Principal advisor with co-advising at 50% with Prof. F. Labeau (McGill University, Canada) and collaborating with Prof. Francisco Messina (Universidad de Buenos Aires, Argentina).

7. **Dr. Malik Boudiaf (Co-Founder - Equall, Canada)**, *Ph.D from Ecole de Technologie Supérieure (ETS)*, Topic: "Information measures for low-shot learning", Graduated in 2023
Co-advising at 50% with Prof. Ismail Ben Ayed (ÉTS, Canada).

8. **Dr. Federica Granese (Junior Researcher at INRIA)**, *Ph.D (co-tutelle) from Institut Polytechnique de Paris and Università degli Studi di Roma*, Topic: "Information-Theoretic Methods for Safety AI", Graduated in 2023
Co-advising at 25% with Dr. Catuscia Palamidessi (LIX, Inria, France) and Daniele Gorla (Università degli Studi di Roma, Italy).

9. **Dr. Mohammad Hadi Shateri (Assistant Professor at ÉTS Montreal)**, *Ph.D from McGill University*, Topic: "Deep Learning and Privacy in Smart Grids", Graduated in 2022
Co-advising at 50% with Prof. F. Labeau (McGill University, Canada) and Prof. Francisco Messina (Universidad de Buenos Aires, Argentina).

10. **Dr. Matias Vera (Research Scientist at Conicet)**, *Ph.D from Universidad de Buenos Aires (Argentina)*, Topic: "Learning via Information-Bottleneck Methods", Graduated in 2020
Co-advised at 50% with Prof. Leonardo Rey Vega (UBA - CONICET, Argentina).

11. **Dr. Clément Feutry (Project Manager - Groupe BPCE)**, *Ph.D from Université Paris-Saclay*, Topic: "Two Sides of Relevant Information: Anonymized Representation through Deep Learning and Predictor Monitoring", Graduated in 2019
Co-advising at 70% with Dr. P. Duhamel (CNRS).

12. **Dr. Kai Wan (Assistant Professor at Huazhong University of Science and Technology)**, *Ph.D from Université Paris-Saclay*, Topic: "Fundamental Limits of Index Coding", Graduated in 2018
Principal advisor with co-advising at 50% with Prof. D. Tuninetti (University of Chicago, US).

13. **Dr. Georg Pichler (Assistant Professor at TU Wien)**, *Ph.D from TU Wien (Austria), Topic: "Information-Theoretic Clustering"*, TUWien, Austria, Graduated in 2017
Co-advised at 50% with Prof. G. Matz (TUWien, Austria).

14. **Dr. Gil Katz (Co-founder | Innovation enthusiast | Mentor @TenGiving)**, *Ph.D from Université Paris-Saclay (France)*, Topic: "Distributed Statistical Inference with Communication Constraints", Graduated in 2017
Co-advising at 50% with Prof. M. Debbah (CentraleSupélec).

15. **Dr. Maggie Mhanna (AI Field Solutions Architect at Google)**, *Ph.D from Université Paris-Saclay* , Topic: "Distributed Hypothesis Testing with Privacy Constraints", Graduated in 2017
Co-advising at 50% with Dr. P. Duhamel (CNRS).

16. **Dr. Chao He (Researcher at Ericsson in Stockholm)**, *Ph.D from Université Paris-Saclay*, Topic: "Cooperation and Feedback in Wireless Networks", Graduated in 2016
Co-advising at 30% with Prof. S.Yang (CentraleSupélec).

17. **Dr. Meryem Benammar (Associate Professor at ISAE SUPAERO)**, *Ph.D from Supélec*, Topic: "Broadcast Channels: Side Information and Security Constraints", Graduated in 2016
Principal advisor.

18. **Dr. German Bassi (Researcher at Ericsson Research)**, *Ph.D from Supélec*, Topic: "Interference Relay Networks: Feedback and Security Constraints", Graduated in 2015
Co-advised at 50% with Prof. S. Yang (CentraleSupélec).

19. **Dr. Andres Altieri (Staff engineer at Qualcomm)**, *Ph.D (co-tutelle) from Supélec and Universidad de Buenos Aires (Argentina)*, Topic:"Fundamental Limits of Cooperative Large Wireless Networks", Graduated in 2014
Co-advised at 50% with Prof. C. Galarza (UBA - CONICET, Argentina) and Prof. Leonardo Rey Vega (UBA - CONICET, Argentina).

20. **Dr. Joffrey Villard (CTO at Syntelier)**, *Ph.D from Supélec*, Topic: "Task-Oriented Source Coding: Secure Transmission, Detection", Graduated in 2012
Co-advised at 50% with Pascal Bianchi (Télécom ParisTech).

21. **Dr. Arash Behboodi (Director of Engineering at Qualcomm AI Research)**, *Ph.D from Supélec*, Topic: "Cooperative Networks with Channel Uncertainty", Graduated in 2012 from Supélec
Advising at 90% (CentraleSupélec).

22. **Dr. Patricia Layec (Team lead & researcher at Nokia Bell Labs)**, *Ph.D from Supélec, CIFRE Orange*, Topic: "Information-Theoretic Bounds and Practical Coding Schemes for MIMO Communications Under Imperfect CSIR and Partial and/or Imperfect CSIT", Graduated in 2009
Co-advising at 20% with Prof. Antoine O. Berthet (CentraleSupélec).

List of Postdoctoral Fellows:

1. **Dr. Andres Altieri (Staff engineer at Qualcomm)**, *"Securing the use of AI"*, June 2023 – May 2024, (L2S, Paris-Saclay University).

2. **Dr. Pierre Colombo (VLM & Multimodal Research @ Cohere)**, *"Securing the use of AI"*, since January 2022 – September 2022, (L2S, Paris-Saclay University).

3. **Dr. Marco Romanelli (Assistant Professor at Hofstra University)**, *"Securing the use of AI"*, since Nov. 2020 – September 2022, (L2S, Paris-Saclay University).

4. **Dr. Hamid Jalalzai**, *"Bounding Information Leakage in Machine Learning"*, February 2021 – September 2022, In collaboration with Dr. Catuscia Palamidessi (LIX, Inria, France).

5. **Dr. Matias Vera (Researcher at CONICET)**, *"Generalization, Information and Learning"*, April 2019 – 2021, In collaboration with Prof. Leonardo Rey Vega (UBA - CONICET, Argentina).

6. **Dr. Francisco Messina (Assistant Professor at UBA)**, *"Artificial Intelligence in Smart Grids: Robust Learning"*, June 2018 - October 2020, In collaboration with Prof. F. Labeau (McGill University, Canada).

7. **Dr. Georg Pichler (Assistant Professor at TU Wien)**, *"Estimation of Information Measures based on Neural Networks"*, July 2018 - Sep. 2019, (L2S, Paris-Saclay University).

8. **Zuxing Li (Assistant Professor at Tongji University)**, *"Distributed Compression and Learning on Power Lines"*, March 2018 - February 2020, In collaboration with RTE France and Prof. M. Kieffer (Paris-Saclay University)
(L2S, Université Paris Saclay).

Selected Honors and Distinctions

2026 - 2030 **Holder of the AI Cluster DATAIA Paris-Saclay - Chaires modulaires**, *Mathematical Principles of Learning and Communication in Attention-Based Models*, Jointly with Prof. Claire Boyer at Laboratoire des Mathématiques d'Orsay (LMO, UPSaclay), 500K Euros

2024 **Area Chair Award**, *Association for Computational Linguistics (ACL 2024)*

2023 **Amazon Research Awards**, *The awards program offers unrestricted funds and AWS Promotional Credits to support research at academic institutions and non-profit organizations in areas of AI*, (website)

2022 **AAAI Outstanding Student Paper Award**, InfoLM: A New Metric to Evaluate Summarization & Data2Text Generation, AAAI Conference on Artificial Intelligence

Jan. 2018 **H2020-MSCA-IF-2017 - Marie Sklodowska-Curie Global Fellowship**, *Information Theory beyond Communications: Distributed Representations and Deep Learning*, European Commission (7% of acceptance rate), 2 years

2016 **IEEE Senior Member**

April 2016 **Visiting Professor Fellowship**, *Information Processing Group, Ecole Polytechnique Federale Lausanne (EPFL)*, Lausanne, Switzerland, 1 week

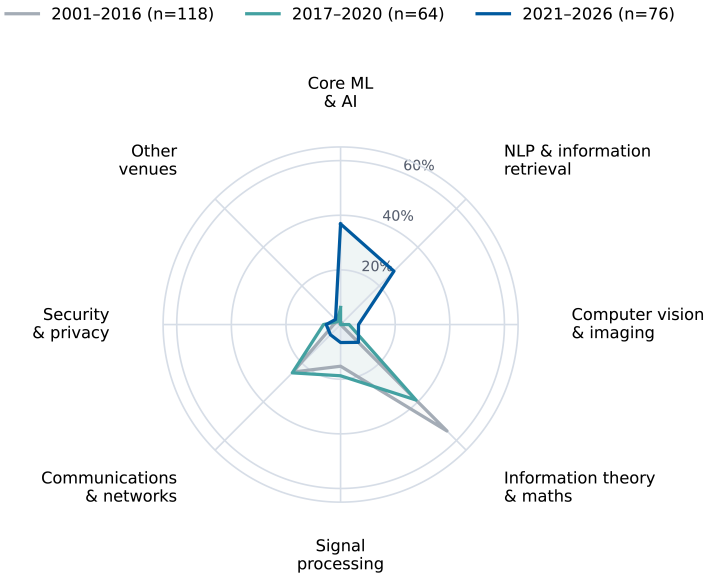
2014 **Coauthor of Best Paper Award**, *Young Researcher Award from Newcom#*

2003 – 2005 **Saint-Exupéry Fellowship**, *French Embassy in Argentina*, France

2001 – 2002 **Merit Award Scholarship**, *University of Buenos Aires*, Buenos Aires, Argentina

Publication Record

Research trajectory. Mathematical and information-theoretic contributions increasingly appear in core machine learning and language research, building on a substantial record in information theory, statistics, and communication theory. This trajectory follows my research on mathematical foundations, evaluation, and reliable AI, with collaborative intelligence as its next frontier. This chart describes publication outlets rather than the boundaries of my expertise.



60 journal articles and **193 international conference papers**, including **20 invited contributions to special sessions**.

Over **50 invited lectures**, including seminars, tutorials, and workshops.

Selected venues.

NeurIPS, ICML, ICLR, AISTATS, AAAI, ACL, EMNLP, CVPR, ECCV, and ACM CCS.

Machine Learning, JMLR, TMLR, IEEE TPAMI, IEEE Transactions on Information Theory, The Annals of Applied Probability, and Information and Inference.

IEEE ISIT, ITW, and ICASSP.

International Scientific Leadership and Service

Coordination of scientific activities (selected subset)

Feb 18 - 20, 2026 **General Chair**, *Workshop: AI for Health*, ETS Montreal, (website)
(up to 110 participants on site)

Oct 1 - 3, 2024 **General Chair**, *Workshop: NLP in the era of generative AI, cognitive sciences, and societal transformation*, Mila - Quebec AI Institute, (website)
(up to 250 participants on site and 3000 online)

May 24 - 26, 2023 **General Co-Chair**, *Workshop ILLS - DATAIA*, Paris-Saclay University, (website)

May 18, 2023 **General Co-Chair**, *Séminaire des IRL CRM et ILLS*, Pôle CRM de l'UQAM, (website)

2021 - 2023 **Conference Committee Chair**, *IEEE Information Theory Society*

July 2019 **General Co-Chair of the 2019**, *IEEE International Symposium on Information Theory (ISIT) in Paris*, IEEE Information Theory Society, (the most prestigious conference in the field with around 1100 participants)
`https://2019.ieee-isit.org`

Nov. 2016 **Conference Special Session**, *Information Theory and Statistical Learning*, Asilomar Conference on Signals, Systems, and Computers

Outreach and participation in scientific committees (selected subset)

2024 - present **Area Chair and Senior Area Chair**, *ICML 2026, NeurIPS 2026, ARR 2026, AAAI 2025-2027, CCS 2025, USENIX 2025, ISIT, and ITW*

2022 - 2023 **International expert for committee evaluation program TRAIL**, *Trusted AI Labs*, Walloon and Brussels regions

2021 - 2022 **Vice-president of the expert committee 23 (Artificial Intelligence) of the ANR for the evaluation of call for research projects**, *ANR*, National Research Agency

2020 - 2021 **Member of the expert committee 23 (Artificial Intelligence) of the ANR for the evaluation of call for research projects**, *ANR*, National Research Agency

Nov. 2020 - 2024 **Board of Entropy**, *Topic Editor of Information Theory, Probability and Statistics*

2019 - 2021 **Associate Editor**, *IEEE Transactions on Information Forensics and Security*

2010 - Present **Session Chair**, *IEEE Intern. Symposium on Information Theory (ISIT) and IEEE Workshop on Information Theory (ITW)*

Service on Ph.D. committees (selected subset)

Dec. 3, 2024 **Ph.D. examiner/president of jury**, *Université Paris-Saclay*, by Alexandre Perez-Label, Évaluation de classifieurs probabilistes et impact sur la décision

Dec. 11th, 2023 **Ph.D. examiner**, *Université Paris-Saclay*, by Fabio Arnez, Deep Neural Network Uncertainty Runtime Monitoring for Robust and Safe AI-based Automated Navigation

Dec. 13th, 2021 **Ph.D. examiner/president of jury**, *Université Paris-Saclay*, by Victor Bouvier, Vers l'apprentissage adaptatif à l'aide de représentations invariantes

Nov. 17th, 2021 **Ph.D. examiner**, *Institut Polytechnique de Paris, Télécom ParisTech*, by Pierre Colombo, Apprendre à représenter et à générer du texte en utilisant des mesures d'information

Nov. 6th, 2020 **Ph.D. reviewer**, *Institut Polytechnique de Paris, Télécom ParisTech*, by Hamid Jalalzai, Learning from Multivariate Extremes: Theory and Application to Natural Language Processing

International and French Collaborations

- At the international level, collaborations have been established with TU Wien (Austria), TU Berlin (Germany), UCLouvain (Belgium), Technion (Israel), McGill (Canada), University of Montreal (Canada), ETS (Canada), INRS (Canada), Universidad de Chile (Chile), Universidad de Buenos Aires (Argentina), University of Chicago (USA), New York University (USA), University of Michigan (USA), among others.
- At the national level, collaborations have been established with Telecom ParisTech, INRIA (Lix - Ecole Polytechnique), Sorbonne université (ISIR), ENS Lyon (LIP), Université Aix-Marseille (LIS), University Paris-Saclay (LMO, LISN), Université de Cergy (ENSEA), among others.

Teaching, Graduate Training and Curriculum Leadership

Courses at Master (M2) Level (selected subset)

2019 – 2020 **Foundations of Privacy and Machine Learning**, *Parisian Master of Research in Computer Science (MPRI)*, ENS Ulm, Ecole Polytechnique, University Paris Diderot, Telecom Paris Tech, University Paris Diderot
(4 lectures of 3h each per year)

2018 – 2021 **Introduction to Deep Learning**, *International summer school on Artificial Intelligence*, Direction International of CentraleSupélec, CentraleSupélec
(2 lectures of 2h30 each per year)

2017 – 2018 **Deep Learning and IoT Data Analytics**, *Master Advanced Wireless Communications Systems (SAR)*, Research master M2, CentraleSupélec
(4 lectures of 3h each per year)

2015 – 2018 **Mathematics of Information and Source Coding**, *Master Multimedia Networking (MN)*, Master M2, Telecom ParisTech/CentraleSupélec/Université Paris-Sud
(8 lectures of 3h each per year)

2009 – 2018 **Elements of Information Theory**, *Master Advanced Wireless Communications Systems (SAR)*, Research master M2, CentraleSupélec
(5 lectures of 3h each per year)

2009 – 2016 **Network Information Theory**, *Master Advanced Wireless Communications Systems (SAR)*, Research Master M2, CentraleSupélec
(4 lectures of 3h each per year)

Courses/TAs/TPs at Engineering Level (selected subset)

2021 – present **Deep Learning**, 3rd year of engineering program at CentraleSupélec
(24h of lectures per year)

2020 **Research Projects**, 2nd and 3rd year of engineering program at CentraleSupélec (24h per year)

2020 – 2021 **Information Theory, Large Dimensional Data and AI**, 2nd year (SG8) of engineering program at CentraleSupélec
(2 lectures of 3h each per year)

2017 – 2018 **Information Theory and Information Complexity**, *1st year of engineering program at CentraleSupélec*, ST4 - responsible of the course and curriculum design (replaced by R. Combes during sabbatical leave)
(around 18h per year)

Around 200h per year, according to the French system (equivalent TD), until Dec. 2021.

Education and Degrees

May 2019 **Habilitation à Diriger les Recherches (HDR)**, *Université Paris Saclay (May 2019)*, France, (ultimate French university degree to serve as Ph.D supervisor)
Information-Theoretic Methods: From Communication to Decision, Privacy and Learning, at the Heart of Data Sciences

May 2007 **Ph.D in Physics**, *Université Paris Sud*, Orsay, France
Multi-user information theory: State information and imperfect channel knowledge

Dec. 2003 **MSc/BSc Mathematics and Electrical Engineering**, *Faculty of Engineering and Mathematics, University of Buenos Aires*, Buenos Aires, Argentina

Dec. 2000 **BSc Electrical Engineering**, *(with Honors) Faculty of Engineering, University of Buenos Aires*, Buenos Aires, Argentina

Earlier Research Experience

Feb. 2004 – Sep. 2004 **Visting Researcher**, *Institut für Nachrichtentechnik und Hochfrequenztechnik*, Vienna, Austria, (working on Capacity of Time-Varying channels)

March 2001 – July 2002 **Young Research Associate**, *Institute of Biomedical Engineering, Faculty of Engineering*, Buenos Aires, Argentina, (working on Speech Recognition)

Invited Seminars, Tutorials and Outreach (subset)

1. Can We Trust AI to Do Science? Reasoning, Evaluation, and the Future of Scientific Discovery, Keynote at AI Tech Days - VINCI Groupe, Nanterre, October 5, 2026.
2. Rational Speech Acts as KL Projections: From Pragmatic Reasoning to Multi-Agent Communication, IVADO Regroupement 2 Workshop Day, Montreal, August 2026.
3. Generative AI in the Age of Multi-Agent Reasoning: *Evaluation, coordination, and scientific discovery*, COMMEX annual meeting of Air Liquide - Keynote IA, France, April 13th, 2026.
4. Generative AI and Scientific Discovery, R7+ Meeting - Keynote IA, Ottawa, November 2025.
5. Information-Theoretic Methods for Trustworthy Machine Learning, STARaCom Annual Meeting at McGill University, Montréal, QC, June 6th, 2023.
6. AI and Data Privacy, Keynote at the Symposium on Artificial Intelligence | MIT-France, July 12th to 13th, 2022: (online)

7. Information-Theoretic Methods in Data Sciences: Model Uncertainty, Robustness and Model Drift, Keynote at Journée Statistique et Informatique pour la Science des Données à Paris Saclay, January 26th, 2022: (online)
8. Information Theory for / by Deep Learning, Keynote at Research Days of Labex Digicosme, November 9th, Université Paris-Saclay (virtual talk), 2020: (online)
9. Learning Anonymized Representations with Mutual Information, MLAI 2019 – International Workshop on Machine Learning and Artificial Intelligence, Paris, October 7th-8th, 2019: (online)
10. Les clés de l'IA: le traitement et l'optimisation de l'information, more than 450 participants (undergrads, Ph.D. students, industry, interested public) at ÉTS, Montréal, Canada, October 16th, 2018. Part I: (online) and Part II: (online)
11. Information-Theoretic Biclustering and Distributed Binary Detection, Royal Institute of Technology in Stockholm (KTH), Stockholm, Sweden, Sep. 14th, 2017.
12. Exploring Connections between Information Theory, Machine Learning and Statistics, Institut National de la Recherche Scientifique (INRS), Montreal, Canada, September 1st, 2016.
13. The Compound Broadcast Channel: Recent Advances and Challenges, Information Processing Group, Ecole Polytechnique Federale Lausanne (EPFL), Switzerland, July 23rd, 2015.
14. On the Multiterminal Interactive Lossy Source Coding Problem, Dept. of EE, Imperial College London, UK, September 17th, 2014.
15. A Proof of the Generalized Markov Lemma with Countable Infinite Alphabets, Dept. of Mathematics, University of Hong Kong, Hong Kong, June 3rd, 2014.
16. An Information-Theoretic Approach to Secure Transmission and Storage of Data Sources, Technische Universität Wien (TUW), Vienna, Austria, April 4th, 2014.
17. The Mathematical Nature of Information, Café sciences, Ecole Centrale de Paris, Paris, France, February 27th, 2014: (online)
18. Capacity of Simultaneous State-Dependent Channels with State Information at the Transmitter, EE Colloquium, Israel Institute of Technology, Technion, Haifa, Israel, April 23, 2009.

Full Publication List

Preprints and submitted/revised articles

[P1] P. Formont, M. Darrin, I. B. Ayed, and P. Piantanida, "MolRGen: A Training and Evaluation Setting for De Novo Molecular Generation with Reasonning Models," 2026. [paper] [arXiv]

[P2] M. Dubois, F. Yvon, and P. Piantanida, "How much were you told? measuring external information in peer reviews," 2026. [paper] [arXiv]

[P3] P. Piantanida and G. Pichler, "A theory for rational speech acts as KL projections with powered semantics," 2026, (submitted)

[P4] Y. Bahram, Z. Dehghani, M. Desbos, E. Granger, P. Piantanida, and M. Shateri, "Continuous adversarial meanflow transfer," 2026. [paper] [arXiv]

[P5] C. Chataigner, P. Piantanida, and G. Farnadi, "Position: Auditing Is Not Evaluating; LLM Audit Requires Dynamic, Contextual, Budget-Aware and Reliable Evidence," Feb. 2026, working paper or preprint. [paper]

[P6] L. Estienne, E. Ernst, M. Vera, P. Piantanida, and L. Ferrer, "Ecuas$_n$: A family of metrics for principled evaluation of uncertainty-augmented systems," 2026. [paper] [arXiv]

Journal papers

[J1] E. Aubinais, E. Gassiat, and P. Piantanida, "Fundamental Limits of Membership Inference Attacks on Machine Learning Models," *Journal of Machine Learning Research*, vol. 26, no. 263, pp. 1–54, 2025. [paper] [arXiv]

[J2] Z. Dehghani Tafti, P. Piantanida, and M. Shateri, "A Universal Source-Free Class Unlearning Framework via Synthetic Embeddings," *Transactions on Machine Learning Research*, May 2026. [paper]

[J3] P. Morales-Álvarez, S. Christodoulidis, M. Vakalopoulou, P. Piantanida, and J. Dolz, "BayesAdapter: Enhanced Uncertainty Estimation in CLIP Few-Shot Adaptation," *International Journal of Computer Vision*, vol. 134, no. 2, p. 51, February 2026. [paper] [arXiv]

[J4] L. Estienne, M. Vera, E. Fons, E. Kochkina, P. Piantanida, and L. Ferrer, "Adapting Language Models to Produce Good Class Probabilities for Classification Tasks," *Transactions on Machine Learning Research*, February 2026. [paper]

[J5] D. Gorla, L. Jalouzot, F. Granese, C. Palamidessi, and P. Piantanida, "On Estimating the Strength of Differentially Private Mechanisms in a Black-Box Setting," *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 5, pp. 5494–5507, September 2025. [paper]

[J6] P. Formont, H. Jeannin, P. Piantanida, and I. Ben Ayed, "A Strong Baseline for Molecular Few-Shot Learning," *Transactions on Machine Learning Research*, February 2025. [paper] [arXiv]

[J7] C. Presvôts, M. Kieffer, T. Prevost, P. Panciatici, Z. Li, and P. Piantanida, "Multiple-model coding scheme for electrical signal compression," *Signal Processing*, vol. 236, p. 110037, November 2025. [paper]

[J8] E. D. C. Gomes, F. Alberge, P. Duhamel, and P. Piantanida, "Combine and Conquer: A Meta-Analysis on Data Shift and Out-of-Distribution Detection," *Transactions on Machine Learning Research*, July 2024. [paper] [arXiv]

[J9] M. Shateri, F. Messina, F. Labeau, and P. Piantanida, "Preserving Privacy in GANs Against Membership Inference Attack," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 1728–1743, 2024. [paper] [arXiv]

[J10] G. Pichler, M. Romanelli, L. Rey Vega, and P. Piantanida, "Perfectly Accurate Membership Inference by a Dishonest Central Server in Federated Learning," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 4, pp. 4290–4296, July 2024. [paper] [arXiv]

[J11] M. Vera, L. Rey Vega, and P. Piantanida, "PACMAN: PAC-style bounds accounting for the Mismatch between Accuracy and Negative log-loss," *Information and Inference: A Journal of the IMA*, vol. 13, no. 1, p. iaae002, January 2024. [paper] [arXiv]

[J12] G. Del Grosso, G. Pichler, C. Palamidessi, and P. Piantanida, "Bounding information leakage in machine learning," *Neurocomputing*, vol. 534, pp. 1–17, May 2023. [paper] [arXiv]

[J13] C. Pinzón, C. Palamidessi, P. Piantanida, and F. Valencia, "On the incompatibility of accuracy and equal opportunity," *Machine Learning*, vol. 113, no. 5, pp. 2405–2434, May 2024. [paper]

[J14] M. Vera, L. Rey Vega, and P. Piantanida, "The role of mutual information in variational classifiers," *Machine Learning*, vol. 112, no. 9, pp. 3105–3150, September 2023. [paper] [arXiv]

[J15] M. Picot, F. Granese, G. Staerman, M. Romanelli, F. Messina, P. Piantanida, and P. Colombo, "A Halfspace-Mass Depth-Based Method for Adversarial Attack Detection," *Transactions on Machine Learning Research*, March 2023. [paper]

[J16] M. Vera, L. Rey Vega, and P. Piantanida, "Information flow in Deep Restricted Boltzmann Machines: An analysis of mutual information between inputs and outputs," *Neurocomputing*, vol. 507, pp. 235–246, October 2022. [paper]

[J17] M. Picot, F. Messina, M. Boudiaf, F. Labeau, I. Ben Ayed, and P. Piantanida, "Adversarial Robustness Via Fisher-Rao Regularization," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 3, pp. 2698–2710, March 2023. [paper] [arXiv]

[J18] M. Shateri, F. Messina, P. Piantanida, and F. Labeau, "Privacy-Cost Management in Smart Meters With Mutual-Information-Based Reinforcement Learning," *IEEE Internet of Things Journal*, vol. 9, no. 22, pp. 22 389–22 398, November 2022. [paper] [arXiv]

[J19] J. F. Silva and P. Piantanida, "On Universal D-Semifaithful Coding for Memoryless Sources With Infinite Alphabets," *IEEE Transactions on Information Theory*, vol. 68, no. 4, pp. 2782–2800, April 2022. [paper] [arXiv]

[J20] S. Espinosa, J. F. Silva, and P. Piantanida, "On the Exponential Approximation of Type II Error Probability of Distributed Test of Independence," *IEEE Transactions on Signal and Information Processing over Networks*, vol. 7, pp. 777–790, 2021. [paper] [arXiv]

[J21] ——, "Finite-Length Bounds on Hypothesis Testing Subject to Vanishing Type I Error Restrictions," *IEEE Signal Processing Letters*, vol. 28, pp. 229–233, 2021. [paper] [arXiv]

[J22] G. Pichler, P. Piantanida, and G. Matz, "Distributed information-theoretic clustering," *Information and Inference: A Journal of the IMA*, vol. 11, no. 1, pp. 137–166, March 2022. [paper] [arXiv]

[J23] K. Wan, D. Tuninetti, M. Ji, and P. Piantanida, "Combination Networks With End-User-Caches: Novel Achievable and Converse Bounds Under Uncoded Cache Placement," *IEEE Transactions on Information Theory*, vol. 68, no. 2, pp. 806–827, February 2022. [paper] [arXiv]

[J24] M. Shateri, F. Messina, P. Piantanida, and F. Labeau, "Real-Time Privacy-Preserving Data Release for Smart Meters," *IEEE Transactions on Smart Grid*, vol. 11, no. 6, pp. 5174–5183, November 2020. [paper] [arXiv]

[J25] A. Bunin, Z. Goldfeld, H. H. Permuter, S. Shamai Shitz, P. Cuff, and P. Piantanida, "Key and Message Semantic-Security Over State-Dependent Channels," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 1541–1556, 2020. [paper] [arXiv]

[J26] K. Wan, D. Tuninetti, M. Ji, G. Caire, and P. Piantanida, "Fundamental Limits of Decentralized Data Shuffling," *IEEE Transactions on Information Theory*, vol. 66, no. 6, pp. 3616–3637, June 2020. [paper] [arXiv]

[J27] K. Wan, D. Tuninetti, and P. Piantanida, "An Index Coding Approach to Caching With Uncoded Cache Placement," *IEEE Transactions on Information Theory*, vol. 66, no. 3, pp. 1318–1332, March 2020. [paper] [arXiv]

[J28] E. de Chérisey, S. Guilley, O. Rioul, and P. Piantanida, "Best Information is Most Successful: Mutual Information and Success Rate in Side-Channel Analysis," *IACR Transactions on Cryptographic Hardware and Embedded Systems*, vol. 2019, no. 2, pp. 49–79, February 2019. [paper]

[J29] J. F. Silva and P. Piantanida, "Universal Weak Variable-Length Source Coding on Countably Infinite Alphabets," *IEEE Transactions on Information Theory*, vol. 66, no. 1, pp. 649–668, January 2020. [paper] [arXiv]

[J30] G. Bassi, P. Piantanida, and S. Shamai Shitz, "The Wiretap Channel With Generalized Feedback: Secure Communication and Key Generation," *IEEE Transactions on Information Theory*, vol. 65, no. 4, pp. 2213–2233, April 2019. [paper] [arXiv]

[J31] A. Altieri and P. Piantanida, "Performance Evaluation of Advanced Relaying Protocols in Large Wireless Networks," *IEEE Access*, vol. 7, pp. 22 214–22 226, 2019. [paper] [arXiv]

[J32] M. Benammar, P. Piantanida, and S. Shamai, "On the Compound Broadcast Channel: Multiple Description Coding and Interference Decoding," *IEEE Transactions on Information Theory*, vol. 66, no. 1, pp. 38–64, January 2020. [paper] [arXiv]

[J33] G. Bassi, P. Piantanida, and S. Shamai (Shitz), "The Secret Key Capacity of a Class of Noisy Channels with Correlated Sources," *Entropy*, vol. 21, no. 8, p. 732, July 2019. [paper]

[J34] M. Vera, L. Rey Vega, and P. Piantanida, "Collaborative Information Bottleneck," *IEEE Transactions on Information Theory*, vol. 65, no. 2, pp. 787–815, February 2019. [paper] [arXiv]

[J35] ——, "Compression-Based Regularization With an Application to Multitask Learning," *IEEE Journal of Selected Topics in Signal Processing*, vol. 12, no. 5, pp. 1063–1076, October 2018. [paper] [arXiv]

[J36] G. Pichler, P. Piantanida, and G. Matz, "Dictator functions maximize mutual information," *The Annals of Applied Probability*, vol. 28, no. 5, pp. 3094–3101, October 2018. [paper] [arXiv]

[J37] C. He, S. Yang, and P. Piantanida, "A Novel Transmission Scheme for the K-User Broadcast Channel With Delayed CSIT," *IEEE Transactions on Wireless Communications*, vol. 17, no. 1, pp. 386–399, January 2018. [paper] [arXiv]

[J38] M. Benammar, P. Piantanida, and S. Shamai Shitz, "Capacity Results for the Multicast Cognitive Interference Channel," *IEEE Transactions on Information Theory*, vol. 63, no. 7, pp. 4119–4136, July 2017. [paper] [arXiv]

[J39] G. Katz, P. Piantanida, and M. Debbah, "Distributed Binary Detection With Lossy Data Compression," *IEEE Transactions on Information Theory*, vol. 63, no. 8, pp. 5207–5227, August 2017. [paper] [arXiv]

[J40] L. Rey Vega, P. Piantanida, and A. O. Hero, "The Three-Terminal Interactive Lossy Source Coding Problem," *IEEE Transactions on Information Theory*, vol. 63, no. 1, pp. 532–562, January 2017. [paper] [arXiv]

[J41] C. He, S. Yang, and P. Piantanida, "On the Gaussian Fading Broadcast Relay Channel With Causal State Feedback," *IEEE Transactions on Communications*, vol. 64, no. 7, pp. 2797–2807, July 2016. [paper]

[J42] M. Benammar and P. Piantanida, "Secrecy Capacity Region of Some Classes of Wiretap Broadcast Channels," *IEEE Transactions on Information Theory*, vol. 61, no. 10, pp. 5564–5582, October 2015. [paper] [arXiv]

[J43] J. Hoydis, R. Couillet, and P. Piantanida, "The Second-Order Coding Rate of the MIMO Quasi-Static Rayleigh Fading Channel," *IEEE Transactions on Information Theory*, vol. 61, no. 12, pp. 6591–6622, December 2015. [paper] [arXiv]

[J44] M. Jabi, A. E. Hamss, L. Szczecinski, and P. Piantanida, "Multipacket Hybrid ARQ: Closing Gap to the Ergodic Capacity," *IEEE Transactions on Communications*, vol. 63, no. 12, pp. 5191–5205, December 2015. [paper] [arXiv]

[J45] G. Bassi, P. Piantanida, and S. Yang, "Capacity Bounds for a Class of Interference Relay Channels," *IEEE Transactions on Information Theory*, vol. 61, no. 7, pp. 3698–3721, July 2015. [paper] [arXiv]

[J46] A. Altieri, P. Piantanida, L. Rey Vega, and C. G. Galarza, "On Fundamental Trade-offs of Device-to-Device Communications in Large Wireless Networks," *IEEE Transactions on Wireless Communications*, vol. 14, no. 9, pp. 4958–4971, September 2015. [paper] [arXiv]

[J47] A. Behboodi and P. Piantanida, "Mixed Noisy Network Coding and Cooperative Unicasting in Wireless Networks," *IEEE Transactions on Information Theory*, vol. 61, no. 1, pp. 189–222, January 2015. [paper] [arXiv]

[J48] A. Altieri, L. Rey Vega, P. Piantanida, and C. G. Galarza, "On the Outage Probability of the Full-Duplex Interference-Limited Relay Channel," *IEEE Journal on Selected Areas in Communications*, vol. 32, no. 9, pp. 1765–1777, September 2014. [paper] [arXiv]

[J49] J. Villard, P. Piantanida, and S. Shamai, "Secure Transmission of Sources Over Noisy Channels With Side Information at the Receivers," *IEEE Transactions on Information Theory*, vol. 60, no. 1, pp. 713–739, January 2014. [paper] [arXiv]

[J50] A. Altieri, L. Rey Vega, P. Piantanida, and C. G. Galarza, "Analysis of a Cooperative Strategy for a Large Decentralized Wireless Network," *IEEE/ACM Transactions on Networking*, vol. 22, no. 4, pp. 1039–1051, August 2014. [paper] [arXiv]

[J51] J. Villard and P. Piantanida, "Secure Multiterminal Source Coding With Side Information at the Eavesdropper," *IEEE Transactions on Information Theory*, vol. 59, no. 6, pp. 3668–3692, June 2013. [paper] [arXiv]

[J52] S. Yang, M. Kobayashi, P. Piantanida, and S. Shamai, "Secrecy Degrees of Freedom of MIMO Broadcast Channels With Delayed CSIT," *IEEE Transactions on Information Theory*, vol. 59, no. 9, pp. 5244–5256, September 2013. [paper] [arXiv]

[J53] A. Zaidi, P. Piantanida, and S. Shamai Shitz, "Capacity Region of Cooperative Multiple-Access Channel With States," *IEEE Transactions on Information Theory*, vol. 59, no. 10, pp. 6153–6174, October 2013. [paper] [arXiv]

[J54] A. Zaidi, S. Shamai, P. Piantanida, and L. Vandendorpe, "Bounds on the Capacity of the Relay Channel With Noncausal State at the Source," *IEEE Transactions on Information Theory*, vol. 59, no. 5, pp. 2639–2672, May 2013. [paper] [arXiv]

[J55] A. Behboodi and P. Piantanida, "Cooperative Strategies for Simultaneous and Broadcast Relay Channels," *IEEE Transactions on Information Theory*, vol. 59, no. 3, pp. 1417–1443, March 2013. [paper] [arXiv]

[J56] M. Kobayashi, P. Piantanida, S. Yang, and S. Shamai, "On the Secrecy Degrees of Freedom of the Multiantenna Block Fading Wiretap Channels," *IEEE Transactions on Information Forensics and Security*, vol. 6, no. 3, pp. 703–711, September 2011. [paper]

[J57] P. Piantanida, G. Matz, and P. Duhamel, "Outage Behavior of Discrete Memoryless Channels Under Channel Estimation Errors," *IEEE Transactions on Information Theory*, vol. 55, no. 9, pp. 4221–4239, September 2009. [paper] [arXiv]

[J58] P. Piantanida, S. M.-S. Sadough, and P. Duhamel, "On the Outage Capacity of a Practical Decoder Accounting for Channel Estimation Inaccuracies," *IEEE Transactions on Communications*, vol. 57, no. 5, pp. 1341–1350, May 2009. [paper] [arXiv]

[J59] A. Zaidi, P. Piantanida, and P. Duhamel, "Broadcast- and MAC-Aware Coding Strategies for Multiple User Information Embedding," *IEEE Transactions on Signal Processing*, vol. 55, no. 6, pp. 2974–2992, June 2007. [paper]

[J60] J. P. Piantanida and C. F. Estienne, "A New Estimator Based on Maximum Entropy," *Latin American Applied Research*, vol. 35, no. 2, pp. 143–147, April–June 2005. [paper]

Proceedings of international conferences

[C1] C. Roverato, J. Cotnareanu, P. Piantanida, B. N. Oreshkin, and M. Coates, "DynaPFN: Zero-shot dynamical system forecasting with tabular prior-fitted networks," in *Advances in Neural Information Processing Systems*, 2026

[C2] C. Chataigner, P. Piantanida, and G. Farnadi, "Auditing is not evaluating: LLM audit requires dynamic, contextual, budget-aware and reliable evidence," in *AI4GOOD Workshop at the 40th Conference on Neural Information Processing Systems (NeurIPS 2026)*, 2026. [paper]

[C3] ——, "Auditing is not evaluating: LLM audit requires dynamic, contextual, budget-aware and reliable evidence," in *Advances in Neural Information Processing Systems*, 2026, neurIPS 2026 Position Paper Track. [paper]

[C4] K. Chehbouni, C. Chataigner, P. Ganesh, P. Piantanida, J. C. K. Cheung, and G. Farnadi, "From infrastructure to interface, the AI value chain drives LLM homogenization," in *Advances in Neural Information Processing Systems*, 2026, neurIPS 2026 Position Paper Track

[C5] A. Satouf, G. D'Erasmo, Y. Zong, H. Amadou-Boubacar, P. Piantanida, and B. Piwowarski, "STORM: Stepwise token optimization with reward-guided beam search," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026. [paper] [arXiv]

[C6] L. Fosse, P. Formont, E. Aubinais, and P. Piantanida, "Data-driven estimation of statistical deficiency," in *2026 IEEE Information Theory Workshop (ITW)*, 2026

[C7] E. Aubinais, P. Formont, P. Piantanida, and E. Gassiat, "Membership Inference Risks in Quantized Models: A Theoretical and Empirical Study," in *The 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. [paper] [arXiv]

[C8] G. Pichler, M. Romanelli, and P. Piantanida, "Happiness as a Measure of Fairness," in *The 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. [paper] [arXiv]

[C9] A. Satouf, Y. Zong, H. A. Boubacar, P. Piantanida, and B. Piwowarski, "QueStER: Query Specification for Generative Keyword-Based Retrieval," in *Findings of the Association for Computational Linguistics: EACL 2026*, ser. Findings of ACL, V. Demberg, K. Inui, and L. Marquez, Eds. Association for Computational Linguistics, March 2026, pp. 5957–5968. [paper] [arXiv]

[C10] A. Satouf, G. B. Zenou, B. Piwowarski, H. A. Boubacar, and P. Piantanida, "Rational Retrieval Acts: Leveraging Pragmatic Reasoning to Improve Sparse Retrieval," in *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, N. Ferro, M. Maistro, G. Pasi, O. Alonso, A. Trotman, and S. Verberne, Eds. ACM, 2025, pp. 2674–2678. [paper] [arXiv]

[C11] P. Marza, L. Fillioux, S. Boutaj, K. Mahatha, C. Desrosiers, P. Piantanida, J. Dolz, S. Christodoulidis, and M. Vakalopoulou, "THUNDER: Tile-level Histopathology image UNDERstanding benchmark," in *Advances in Neural Information Processing Systems 38*. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2025, pp. 171 942–172 034, (spotlight). [paper] [arXiv]

[C12] P. Formont, M. DARRIN, B. Karimian, E. Granger, J. C. Cheung, I. B. Ayed, M. Shateri, and P. Piantanida, "Learning Task-Agnostic Representations through Multi-Teacher Distillation," in *Advances in Neural Information Processing Systems 38*. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2025, pp. 121 561–121 607. [paper] [arXiv]

[C13] C. Spinoso-Di Piano, D. E. Austin, P. Piantanida, and J. C. Cheung, "$(RSA)^2$: A Rhetorical-Strategy-Aware Rational Speech Act Framework for Figurative Language Understanding," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, July 2025, pp. 20 898–20 938. [paper] [arXiv]

[C14] L. Estienne, G. B. Zenou, N. Naderi, J. C. Cheung, and P. Piantanida, "Collaborative Rational Speech Act: Pragmatic Reasoning for Multi-Turn Dialog," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, November 2025, pp. 22 509–22 523, (oral). [paper] [arXiv]

[C15] L. Fosse, F. Bechet, B. Favre, G. Damnati, G. Lecorvé, M. Darrin, P. Formont, and P. Piantanida, "Statistical Deficiency for Task Inclusion Estimation," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, July 2025, pp. 382–415. [paper] [arXiv]

[C16] M. Dubois, F. Yvon, and P. Piantanida, "MOSAIC: Multiple Observers Spotting AI Content," in *Findings of the Association for Computational Linguistics: ACL 2025*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, July 2025, pp. 24 230–24 247. [paper] [arXiv]

[C17] ——, "MOSAIC at GENAI Detection Task 3 : Zero-Shot Detection Using an Ensemble of Models," in *Proceedings of the 1stWorkshop on GenAI Content Detection (GenAIDetect)*. International Conference on Computational Linguistics, January 2025, pp. 371–376. [paper]

[C18] O. Beaumont, R. Bourgouin, M. Darrin, L. Marchal, and P. Piantanida, "Leveraging Expert Usage to Speed up LLM Inference with Expert Parallelism," in *Euro-Par 2025: Parallel Processing - 31st European Conference on Parallel and Distributed Processing, Dresden, Germany, August 25-29, 2025, Proceedings, Part I*, ser. Lecture Notes in Computer Science, W. E. Nagel, D. Goehringer, and P. C. Diniz, Eds., vol. 15900. Springer, 2025, pp. 145–158. [paper]

[C19] M. Dubois, F. Yvon, and P. Piantanida, "How Sampling Affects the Detectability of Machine-written texts: A Comprehensive Study," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Suzhou, China: Association for Computational Linguistics, November 2025, pp. 11 369–11 387. [paper] [arXiv]

[C20] C. Presvôts, M. Kieffer, T. Prevost, P. Panciatici, Z. Li, and P. Piantanida, "Two-stage Multiple-Model Compression Approach for Sampled Electrical Signals," in *2024 Data Compression Conference (DCC)*, A. Bilgin, J. E. Fowler, J. Serra-Sagristà, Y. Ye, and J. A. Storer, Eds. IEEE, 2024, pp. 522–531. [paper]

[C21] M. Darrin, P. Formont, I. B. Ayed, J. C. Cheung, and P. Piantanida, "When is an Embedding Model More Promising than Another?" in *Advances in Neural Information Processing Systems*

37. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024, pp. 68 330–68 379. [paper] [arXiv]

[C22] A. Altieri, M. Romanelli, G. Pichler, F. Alberge, and P. Piantanida, "Beyond the Norms: Detecting Prediction Errors in Regression Models," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 21–27 Jul 2024, pp. 1186–1221, (spotlight). [paper] [arXiv]

[C23] E. D. C. Gomes, M. Romanelli, G. Pichler, and P. Piantanida, "A Data-Driven Measure of Relative Uncertainty for Misclassification Detection," in *The Twelfth International Conference on Learning Representations*, 2024. [paper] [arXiv]

[C24] M. Darrin, P. Formont, J. Cheung, and P. Piantanida, "COSMIC: Mutual Information for Task-Agnostic Summarization Evaluation," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, August 2024, pp. 12 696–12 717, (oral). [paper] [arXiv]

[C25] M. Darrin, I. Arous, P. Piantanida, and J. Cheung, "GLIMPSE: Pragmatically Informative Multi-Document Summarization for Scholarly Reviews," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, August 2024, pp. 12 737–12 752. [paper] [arXiv]

[C26] F. Granese, M. Romanelli, and P. Piantanida, "Optimal Zero-Shot Detector for Multi-Armed Attacks," in *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, S. Dasgupta, S. Mandt, and Y. Li, Eds., vol. 238. PMLR, 02–04 May 2024, pp. 2467–2475. [paper] [arXiv]

[C27] D. Gorla, L. Jalouzot, F. Granese, C. Palamidessi, and P. Piantanida, "On the (Im)Possibility of Estimating Various Notions of Differential Privacy (short paper)," in *Proceedings of the 24th Italian Conference on Theoretical Computer Science, Palermo, Italy, September 13-15, 2023*, ser. CEUR Workshop Proceedings, G. Castiglione and M. Sciortino, Eds., vol. 3587. CEUR-WS.org, 2023, pp. 219–224. [paper] [arXiv]

[C28] P. Colombo, M. Picot, N. Noiry, G. Staerman, and P. Piantanida, "Toward Stronger Textual Attack Detectors," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, H. Bouamor, J. Pino, and K. Bali, Eds. Association for Computational Linguistics, 2023, pp. 484–505. [paper] [arXiv]

[C29] M. Darrin, G. Staerman, E. D. C. Gomes, J. C. Cheung, P. Piantanida, and P. Colombo, "Unsupervised Layer-wise Score Aggregation for Textual OOD Detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38. Association for the Advancement of Artificial Intelligence (AAAI), February 2024, pp. 17 880–17 888. [paper] [arXiv]

[C30] P. Colombo, M. Peyrard, N. Noiry, R. West, and P. Piantanida, "The Glass Ceiling of Automatic Evaluation in Natural Language Generation," in *Findings of the Association for Computational Linguistics: IJCNLP-AACL 2023 (Findings)*, J. C. Park, Y. Arase, B. Hu, W. Lu, D. Wijaya, A. Purwarianti, and A. A. Krisnadhi, Eds. Nusa Dua, Bali: Association for Computational Linguistics, November 2023, pp. 178–183. [paper] [arXiv]

[C31] M. Darrin, P. Piantanida, and P. Colombo, "RainProof: An Umbrella to Shield Text Generator from Out-Of-Distribution Data," in *Proceedings of the 2023 Conference on Empirical*

Methods in Natural Language Processing, H. Bouamor, J. Pino, and K. Bali, Eds. Singapore: Association for Computational Linguistics, December 2023, pp. 5831–5857. [paper] [arXiv]

[C32] P. Colombo, N. Noiry, G. Staerman, and P. Piantanida, "A Novel Information Theoretic Objective to Disentangle Representations for Fair Classification," in *Findings of the Association for Computational Linguistics: IJCNLP-AACL 2023 (Findings)*, J. C. Park, Y. Arase, B. Hu, W. Lu, D. Wijaya, A. Purwarianti, and A. A. Krisnadhi, Eds. Nusa Dua, Bali: Association for Computational Linguistics, November 2023, pp. 184–198. [paper] [arXiv]

[C33] P. Colombo, V. Pellegrain, M. Boudiaf, M. Tami, V. Storchan, I. Ayed, and P. Piantanida, "Transductive Learning for Textual Few-Shot Classification in API-based Embedding Models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, H. Bouamor, J. Pino, and K. Bali, Eds. Singapore: Association for Computational Linguistics, December 2023, pp. 4214–4231. [paper] [arXiv]

[C34] M. Boudiaf, E. Bennequin, M. Tami, C. Hudelot, A. Toubhans, P. Piantanida, and I. B. Ayed, "Open-Set Likelihood Maximization for Few-Shot Learning," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2023, pp. 24 007–24 016. [paper] [arXiv]

[C35] P. Colombo, E. Dadalto, G. Staerman, N. Noiry, and P. Piantanida, "Beyond Mahalanobis Distance for Textual OOD Detection," in *Advances in Neural Information Processing Systems 35*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, Eds., vol. 35. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2022, pp. 17 744–17 759. [paper] [arXiv]

[C36] N. M. Guerreiro, P. Colombo, P. Piantanida, and A. Martins, "Optimal Transport for Unsupervised Hallucination Detection in Neural Machine Translation," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 13 766–13 784. [paper] [arXiv]

[C37] F. Granese, M. Picot, M. Romanelli, F. Messina, and P. Piantanida, "MEAD: A Multi-Armed Approach for Evaluation of Adversarial Examples Detectors," in *Machine Learning and Knowledge Discovery in Databases - European Conference, ECML PKDD 2022, Grenoble, France, September 19-23, 2022, Proceedings, Part III*, ser. Lecture Notes in Computer Science, M. Amini, S. Canu, A. Fischer, T. Guns, P. K. Novak, and G. Tsoumakas, Eds., vol. 13715. Springer Nature Switzerland, 2022, pp. 286–303. [paper] [arXiv]

[C38] G. Pichler, P. J. A. Colombo, M. Boudiaf, G. Koliander, and P. Piantanida, "A Differential Entropy Estimator for Training Neural Networks," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 17–23 Jul 2022, pp. 17 691–17 715. [paper] [arXiv]

[C39] P. Colombo, G. Staerman, N. Noiry, and P. Piantanida, "Learning Disentangled Textual Representations via Statistical Measures of Similarity," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 2614–2630, (spotlight). [paper] [arXiv]

[C40] G. Del Grosso, H. Jalalzai, G. Pichler, C. Palamidessi, and P. Piantanida, "Leveraging Adversarial Examples to Quantify Membership Information Leakage," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, Jun 19, 2022 - Jun 24, 2022, New Orleans, Louisiana, Nov 16 2022, pp. 10 389–10 399. [paper] [arXiv]

[C41] S. Espinosa, J. F. Silva, and P. Piantanida, "A Data-Driven Quantization Design for Distributed Testing Against Independence with Communication Constraints," in *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 22-27 May 2022 in Singapore. 2022, pp. 5238–5242. [paper]

[C42] E. Dadalto Carama Gome, F. Alberge, P. Duhamel, and P. Piantanida, "Igeood: An Information Geometry Approach to Out-of-Distribution Detection ," in *The Tenth International Conference on Learning Representations*, (Virtual) Mon Apr 25th through Fri the 29th 2022. [paper] [arXiv]

[C43] P. Colombo, C. Clavel, and P. Piantanida, "InfoLM: A New Metric to Evaluate Summarization & Data2Text Generation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36. Association for the Advancement of Artificial Intelligence (AAAI), Feb. 22 - March 1 2022, pp. 10 554–10 562, doi:https://doi.org/10.1609/aaai.v36i10.21299 (Best Paper Award). [paper] [arXiv]

[C44] C. Pinzon, C. Palamidessi, P. Piantanida, and F. Valencia, "On the Impossibility of non-Trivial Accuracy in Presence of Fairness Constraints," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36. Association for the Advancement of Artificial Intelligence (AAAI), Feb. 22 - March 1 2022, pp. 7993–8000. [paper] [arXiv]

[C45] M. Picot, F. Messina, F. Labeau, and P. Piantanida, "Robust Autoencoder-based State Estimation in Power Systems," in *IEEE ISGT NA 2022: "Technology Solutions for an Evolving Grid"*, February 21 - 24, Washington, DC, USA 2022

[C46] G. Del Grosso, G. Pichler, C. Palamidessi, and P. Piantanida, "Formalizing Attribute and Membership Inference Attacks on Machine Learning Models," in *Privacy Preserving Machine Learning*, Nov. 19th, Virtual ACM CCS 2021 Workshop 2021. [paper]

[C47] E. Dadalto Carama Gome, F. Alberge, P. Duhamel, and P. Piantanida, "Igeood: An Information Geometry Approach to Out-of-Distribution Detection ," in *NeurIPS 2021 Workshop on Distribution Shifts*, Dec 13th, 2021, Virtual 2021

[C48] P. Colombo, G. Staerman, C. Clavel, and P. Piantanida, "Automatic Text Evaluation through the Lens of Wasserstein Barycenters," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, November 2021, pp. 10 450–10 466. [paper] [arXiv]

[C49] F. Granese, M. Romanelli, D. Gorla, C. Palamidessi, and P. Piantanida, "DOCTOR: A Simple Method for Detecting Misclassification Errors," in *Advances in Neural Information Processing Systems*, Dec 6th - 14th, 2021, Virtual 2021, (spotlight). [paper] [arXiv]

[C50] O. Veilleux, M. Boudiaf, P. Piantanida, and I. Ben Ayed, "Realistic evaluation of transductive few-shot learning," in *Advances in Neural Information Processing Systems*, Dec 6th - 14th, 2021, Virtual 2021. [paper] [arXiv]

[C51] M. Benammar, E. Dadalto Carama Gome, and P. Piantanida, "CSI-aided Robust Neural-based Decoders," in *The 11th International Symposium on Topics in Coding*, Virtual August 30th to September 3rd 2021

[C52] P. Colombo, P. Piantanida, and C. Clavel, "A Novel Estimator of Mutual Information for Learning to Disentangle Textual Representations," in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Association for Computational Linguistics, August 2021, pp. 6539–6550, (spotlight). [paper] [arXiv]

[C53] M. Boudiaf, H. Kervadec, I. M. Ziko, P. Piantanida, I. Ben Ayed, and J. Dolz, "Few-Shot Segmentation Without Meta-Learning: A Good Transductive Inference Is All You Need?" in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, Virtual Sat, Jun 19, 2021 – Fri, Jun 25, 2021 2021, pp. 13 974–13 983. [paper] [arXiv]

[C54] M. Vera, L. Rey Vega, and P. Piantanida, "Information and Regularization in Restricted Boltzmann Machines," in *ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Sun, Jun 6, 2021 – Fri, Jun 11 2021, pp. 3155–3159. [paper]

[C55] M. Shateri, F. Messina, P. Piantanida, and F. Labeau, "Learning Sparse Privacy-Preserving Representations for Smart Meters Data," in *EEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (IEEE SmartGridComm 2021)*, Oct. 25 - Oct. 28 2021. [paper] [arXiv]

[C56] M. Boudiaf, J. Rony, I. M. Ziko, E. Granger, M. Pedersoli, P. Piantanida, and I. Ben Ayed, "A Unifying Mutual Information View of Metric Learning: Cross-Entropy vs. Pairwise Losses," in *European Conference on Computer Vision (ECCV)*, 2020, pp. 548–564, (spotlight). [paper] [arXiv]

[C57] M. Boudiaf, I. Ziko, J. Rony, J. Dolz, I. Ben Ayed, and P. Piantanida, "Information Maximization for Few-Shot Learning," in *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, and H. Lin, Eds., vol. 33. Curran Associates, Inc., Dec 5th - 12th, 2020, Vancouver 2020, pp. 2445–2457. [paper] [arXiv]

[C58] M. Romanelli, K. Chatzikokolakis, C. Palamidessi, and P. Piantanida, "Estimating G-Leakage via Machine Learning," in *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS 2020. New York, NY, USA: ACM, 2020, pp. 697–716. [paper] [arXiv]

[C59] G. Del Grosso, G. Pichler, and P. Piantanida, "Privacy-Preserving Synthetic Smart Meters Data," in *2021 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*. IEEE, February 15-18, Washington, DC, USA 2021, pp. 1–5. [paper] [arXiv]

[C60] M. Shateri, F. Messina, P. Piantanida, and F. Labeau, "On the Impact of Side Information on Smart Meter Privacy-Preserving Methods," in *2020 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*. IEEE, October 2020, pp. 1–6. [paper] [arXiv]

[C61] G. Pichler, J. Dolz, I. Ben Ayed, and P. Piantanida, "On Direct Distribution Matching for Adapting Segmentation Networks," in *Proceedings of the Third Conference on Medical Imaging with Deep Learning*, ser. Proceedings of Machine Learning Research, vol. 121. PMLR, 06–08 Jul 2020, pp. 624–637. [paper] [arXiv]

[C62] M. Shateri, F. Messina, P. Piantanida, and F. Labeau, "Privacy-Cost Management in Smart Meters Using Deep Reinforcement Learning," in *2020 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe)*. IEEE, 2020, pp. 929–933. [paper] [arXiv]

[C63] C. Feutry, P. Piantanida, and P. Duhamel, "Learning Semi-supervised Anonymized Representations by Mutual Information," in *ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, May 4 and May 8, Barcelona, Spain 2020, pp. 3467–3471. [paper]

[C64] M. Shateri, F. Messina, P. Piantanida, and F. Labeau, "Recurrent Adversarial Training for Smart Meter Privacy," in *Montreal AI Symposium*, August 2019

[C65] ——, "Deep Directed Information-Based Learning for Privacy-Preserving Smart Meter Data Release," in *2019 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*. IEEE, October 2019, pp. 1–7. [paper] [arXiv]

[C66] G. Pichler, J. Dolz, I. Ben Ayed, and P. Piantanida, "On Direct Distribution Matching for Adapting Segmentation Networks," in *Montreal AI Symposium*, August 2019

[C67] S. Espinosa, J. F. Silva, and P. Piantanida, "New Results on Testing Against Independence with Rate-Limited Constraints," in *2019 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*. IEEE, 2019, pp. 1–5. [paper]

[C68] E. de Chérisey, S. Guilley, O. Rioul, and P. Piantanida, "An Information-Theoretic Model for Side-Channel Attacks in Embedded Hardware," in *2019 IEEE International Symposium on Information Theory (ISIT)*. IEEE, July 7-12 2019, pp. 310–315. [paper]

[C69] J. Silva and P. Piantanida, "Universal D-Semifaithfull Coding for Countably Infinite Alphabets," in *2019 IEEE International Symposium on Information Theory (ISIT)*. IEEE, July 7-12 2019, pp. 567–571. [paper]

[C70] K. Wan, D. Tuninetti, M. Ji, and P. Piantanida, "A Novel Asymmetric Coded Placement in Combination Networks with End-User Caches," in *2018 Information Theory and Applications Workshop (ITA)*. IEEE, Feb 2018, pp. 1–5. [paper] [arXiv]

[C71] ——, "Fundamental Limits of Distributed Data Shuffling," in *2018 56th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, Oct 2018, pp. 662–669. [paper]

[C72] F. Alberge, C. Feutry, P. Duhamel, and P. Piantanida, "Detecting Covariate Shift with Black Box Predictors," in *2019 26th International Conference on Telecommunications (ICT)*. IEEE, April 2019, pp. 324–329. [paper]

[C73] L. Dogariu, S. Ciochina, C. Paleologu, J. Benesty, and P. Piantanida, "An Optimized LMS Algorithm for Bilinear Forms," in *2018 International Symposium on Electronics and Telecommunications (ISETC)*. IEEE, Nov 2018, pp. 1–4. [paper]

[C74] C. Feutry, P. Piantanida, Y. Bengio, and P. Duhamel, "Learning Anonymized Representations with Adversarial Neural Networks," in *2018 International Conference on Machine Learning, Stockholm, Sweden*, Jul 10, 2018 – Jul 15 2018. [paper] [arXiv]

[C75] M. Vera, P. Piantanida, and L. R. Vega, "The Role of the Information Bottleneck in Representation Learning," in *2018 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2018, pp. 1580–1584. [paper]

[C76] K. Wan, M. Ji, P. Piantanida, and D. Tuninetti, "On the Benefits of Asymmetric Coded Cache Placement in Combination Networks with End-User Caches," in *2018 IEEE International Symposium on Information Theory (ISIT)*, June 2018, pp. 1550–1554. [paper]

[C77] ——, "Caching in Combination Networks: Novel Multicast Message Generation and Delivery by Leveraging the Network Topology," in *2018 IEEE International Conference on Communications (ICC)*. IEEE, May 2018, pp. 1–6. [paper] [arXiv]

[C78] G. Bassi, M. Skoglund, and P. Piantanida, "Lossy Communication Subject to Statistical Parameter Privacy," in *2018 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2018, pp. 1031–1035. [paper]

[C79] A. Bunin, Z. Goldfeld, H. H. Permuter, S. Shamai, P. Cuff, and P. Piantanida, "Key-Message Security over State-Dependent Wiretap Channels," in *2018 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2018, pp. 136–140. [paper]

[C80] L. M. Dogariu, S. Ciochin, C. Paleologu, J. Benesty, and P. Piantanida, "Identification of Bilinear Forms with the Kalman Filter," in *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, April 2018, pp. 4134–4138. [paper]

[C81] M. Vera, P. Piantanida, and L. Rey Vega, "A learning algorithm with compression-based regularization," in *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, April 2018, pp. 2836–2840. [paper]

[C82] K. Wan, M. Ji, P. Piantanida, and D. Tuninetti, "Caching in Combination Networks: Novel Multicast Message Generation and Delivery by Leveraging the Network Topology," in *IEEE ICC 2018 Communication Theory Symposium, 20 – 24 May, Kansas City, MO, USA*, 2018. [paper] [arXiv]

[C83] ——, "Novel Outer Bounds for Combination Networks with End-User-Caches," in *2017 IEEE Information Theory Workshop (ITW)*. IEEE, 2017, pp. 444–448. [paper]

[C84] Q. Yang, P. Piantanida, and D. Gunduz, " The Multi-Layer Information Bottleneck Problem," in *Information Theory Workshop (ITW), 2017 IEEE, Nov. 6th – 10th*, 2017. [paper] [arXiv]

[C85] L. M. Dogariu, S. Ciochin, C. Paleologu, J. Benesty, and P. Piantanida, "An adaptive solution for nonlinear system identification," in *2017 International Symposium on Signals, Circuits and Systems (ISSCS)*. IEEE, July 2017, pp. 1–4. [paper]

[C86] C. He, S. Yang, and P. Piantanida, "A non-linear transmission scheme for the K-user broadcast channel with state feedback," in *2016 International Symposium on Information Theory and Its Applications (ISITA)*, Oct 2016, pp. 101–105

[C87] J. F. Silva and P. Piantanida, "The redundancy gains of almost lossless universal source coding over envelope families," in *2017 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2017, pp. 2003–2007. [paper]

[C88] M. Vera, L. R. Vega, and P. Piantanida, "Distributed cooperative information bottleneck," in *2017 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2017, pp. 709–713. [paper]

[C89] G. Pichler, P. Piantanida, and G. Matz, "A multiple description CEO problem with log-loss distortion," in *2017 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2017, pp. 111–115. [paper]

[C90] M. Mhanna, P. Piantanida, and P. Duhamel, "Privacy-preserving quantization learning with applications to smart meters," in *2017 IEEE International Conference on Communications (ICC)*. IEEE, May 2017, pp. 1–6. [paper]

[C91] ——, "Privacy-preserving quantization learning for distributed binary decision with applications to smart meters," in *2017 IEEE International Conference on Communications Workshops (ICC Workshops)*. IEEE, May 2017, pp. 1128–1133. [paper]

[C92] K. Wan, D. Tuninetti, and P. Piantanida, "Novel delivery schemes for decentralized coded caching in the finite file size regime," in *2017 IEEE International Conference on Communications Workshops (ICC Workshops)*. IEEE, May 2017, pp. 1183–1188. [paper] [arXiv]

[C93] K. Wan, M. Ji, P. Piantanida, and D. Tuninetti, "Novel inner bounds with uncoded cache placement for combination networks with end-user-caches," in *2017 55th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, Oct 2017, pp. 361–368. [paper]

[C94] G. Katz, P. Piantanida, and M. Debbah, "A New Approach to Distributed Hypothesis Testing," in *2016 50th Asilomar Conference on Signals, Systems and Computers*. IEEE, Nov. 2016, pp. 1365–1369. [paper]

[C95] A. Bunin, P. Piantanida, and S. S. Shitz, "The Gaussian wiretap channel with correlated sources at the terminals: Secret communication and key generation," in *2016 IEEE International Conference on the Science of Electrical Engineering (ICSEE)*. IEEE, Nov 2016, pp. 1–5. [paper]

[C96] J. F. Silva and P. Piantanida, "Almost lossless variable-length source coding on countably infinite alphabets," in *2016 IEEE International Symposium on Information Theory (ISIT)*. IEEE, July 2016, pp. 1–5. [paper]

[C97] G. Pichler, P. Piantanida, and G. Matz, "Distributed information-theoretic biclustering," in *2016 IEEE International Symposium on Information Theory (ISIT)*. IEEE, July 2016, pp. 1083–1087. [paper] [arXiv]

[C98] G. Bassi, P. Piantanida, and S. Shamai, "Secret key generation over noisy channels with common randomness," in *2016 IEEE International Symposium on Information Theory (ISIT)*. IEEE, July 2016, pp. 510–514. [paper]

[C99] G. Katz, P. Piantanida, and M. Debbah, "Collaborative distributed hypothesis testing with general hypotheses," in *2016 IEEE International Symposium on Information Theory (ISIT)*. IEEE, July 2016, pp. 1705–1709. [paper]

[C100] K. Wan, D. Tuninetti, and P. Piantanida, "On caching with more users than files," in *2016 IEEE International Symposium on Information Theory (ISIT)*. IEEE, July 2016, pp. 135–139. [paper] [arXiv]

[C101] G. Pichler, P. Piantanida, and G. Matz, "Tight Upper Bound on the Mutual Information of Two Boolean Functions," in *2016 IEEE Information Theory Workshop (ITW)*. IEEE, 2016, pp. 16–20. [paper]

[C102] K.Wan, D. Tuninetti, and P. Piantanida, "On the Optimality of Uncoded Cache Placement," in *2016 IEEE Information Theory Workshop (ITW)*. IEEE, 2016, pp. 161–165. [paper] [arXiv]

[C103] M. Mhanna, P. Duhamel, and P. Piantanida, "Quantization for distributed binary detection under secrecy constraints," in *2016 IEEE International Conference on Communications (ICC)*. IEEE, May 2016, pp. 1–6. [paper]

[C104] G. Pichler, P. Piantanida, and G. Matz, "Distributed information-theoretic biclustering of two memoryless sources," in *2015 53rd Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, Sept 2015, pp. 426–433. [paper]

[C105] G. Bassi, P. Piantanida, and S. Shamai, "The Wiretap Channel with Generalized Feedback: Secure Communication and Key Generation," in *2015 IEEE Information Theory Workshop - Fall (ITW)*. IEEE, October 2015, pp. 282–286. [paper] [arXiv]

[C106] M. Vera, L. Rey Vega, and P. Piantanida, "The two-way cooperative Information Bottleneck," in *2015 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2015, pp. 2131–2135. [paper]

[C107] L. Rey Vega, P. Piantanida, and A. O. Hero, "On the rate-distortion regions for interactive source coding," in *2015 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2015, pp. 1771–1775. [paper]

[C108] G. Bassi, P. Piantanida, and S. Shamai, "On the capacity of the wiretap channel with generalized feedback," in *2015 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2015, pp. 1154–1158. [paper]

[C109] M. Benammar, P. Piantanida, and S. Shamai, "On Multiple Description coding for the Multicast Cognitive Interference Channel," in *2015 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2015, pp. 2126–2130. [paper]

[C110] M. Mhanna and P. Piantanida, "On secure distributed hypothesis testing," in *2015 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2015, pp. 1605–1609. [paper]

[C111] G. Katz, P. Piantanida, R. Couillet, and M. Debbah, "On the necessity of binning for the distributed hypothesis testing problem," in *2015 IEEE International Symposium on Information Theory (ISIT)*. IEEE, June 2015, pp. 2797–2801. [paper]

[C112] M. Benammar, P. Piantanida, and S. Shamai, "Capacity results for the Multicast Cognitive Interference Channel," in *2015 IEEE Information Theory Workshop (ITW)*. IEEE, April 2015, pp. 1–5. [paper] [arXiv]

[C113] C. He, S. Yang, and P. Piantanida, "An achievable rate region of broadcast relay channel with state feedback," in *2015 IEEE International Conference on Communications (ICC)*. IEEE, June 2015, pp. 4193–4198. [paper]

[C114] A. El Hamss, L. Szczecinski, and P. Piantanida, "Increasing the throughput of HARQ via multi-packet transmission," in *2014 IEEE Global Communications Conference*. IEEE, Dec 2014, pp. 1485–1491. [paper]

[C115] G. Katz, P. Piantanida, R. Couillet, and M. Debbah, "Joint estimation and detection against independence," in *2014 52nd Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, Sept 2014, pp. 1220–1227. [paper]

[C116] M. Benammar, P. Piantanida, and S. Shamai (Shitz), "Multiple Description Coding for the Compound Broadcast Channel," in *2014 IEEE Information Theory Workshop (ITW 2014)*. IEEE, November 2014, pp. 566–570. [paper]

[C117] M. Benammar and P. Piantanida, "On the secrecy capacity region of a class of parallel Wiretap Broadcast Channel," in *2014 52nd Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, Sept 2014, pp. 1072–1077. [paper]

[C118] ——, "On the secrecy capacity region of the Wiretap Broadcast Channel," in *2014 IEEE Information Theory Workshop (ITW 2014)*. IEEE, Nov 2014, pp. 421–425. [paper]

[C119] G. Bassi, P. Piantanida, and S. Yang, "Constant-gap results and cooperative strategies for a class of Interference Relay Channels," in *2014 IEEE International Symposium on Information Theory*. IEEE, June 2014, pp. 1421–1425. [paper]

[C120] P. Piantanida, L. Vega, and A. Hero, "A proof of the Generalized Markov Lemma with countable infinite sources," in *2014 IEEE International Symposium on Information Theory*. IEEE, June 2014, pp. 591–595. [paper]

[C121] R. Tandon, P. Piantanida, and S. Shamai, "On multi-user MISO wiretap channels with delayed CSIT," in *2014 IEEE International Symposium on Information Theory*. IEEE, June 2014, pp. 211–215. [paper]

[C122] L. Rey Vega, P. Piantanida, and A. Hero, "On the three-terminal interactive lossy source coding problem," in *2014 IEEE International Symposium on Information Theory*. IEEE, June 2014, pp. 771–775. [paper] [arXiv]

[C123] G. Bassi, P. Piantanida, and S. Yang, "Capacity to within a Constant Gap for a Class of Interference Relay Channels," in *2013 51st Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, 2013, pp. 1300–1306. [paper]

[C124] P. Piantanida, M. Kobayashi, and G. Caire, "Analog Index Coding over Block-Fading MISO Broadcast Channels with Feedback," in *2013 IEEE Information Theory Workshop (ITW)*. IEEE, 2013, pp. 1–5. [paper]

[C125] M. Benammar and P. Piantanida, "On the Role of Interference Decoding in Compound Broadcast Channels," in *2013 IEEE Information Theory Workshop (ITW)*. IEEE, 2013, pp. 1–5. [paper]

[C126] J. Hoydis, R. Couillet, and P. Piantanida, "Bounds on the Second-Order Coding Rate of the MIMO Rayleigh Block-Fading Channel," in *2013 IEEE International Symposium on Information Theory*. IEEE, 2013, pp. 1526–1530. [paper] [arXiv]

[C127] A. Altieri, L. Vega, C. Galarza, and P. Piantanida, "Cooperative Unicasting in Large Wireless Networks ," in *2013 IEEE International Symposium on Information Theory*. IEEE, 2013, pp. 429–433. [paper]

[C128] A. Behboodi and P. Piantanida, "Mixed Noisy Network Coding ," in *2013 IEEE International Symposium on Information Theory*. IEEE, 2013, pp. 2324–2328. [paper]

[C129] J. Hoydis, R. Couillet, P. Piantanida, and M. Debbah, "A random matrix approach to the finite blocklength regime of MIMO fading channels," in *2012 IEEE International Symposium on Information Theory Proceedings*. IEEE, july 2012, pp. 2181–2185. [paper]

[C130] A. Altieri, L. Vega, C. Galarza, and P. Piantanida, "Cooperation versus interference in large wireless relay networks," in *2012 IEEE International Symposium on Information Theory Proceedings*. IEEE, july 2012, pp. 611–615. [paper]

[C131] M. Ivashkina, I. Andriyanova, P. Piantanida, and C. Poulliat, "Erasure-correcting vs. erasure-detecting codes for the full-duplex binary erasure relay channel," in *2012 IEEE International Symposium on Information Theory Proceedings*. IEEE, july 2012, pp. 945–949. [paper]

[C132] A. Behboodi and P. Piantanida, "Selective coding strategy for unicast composite networks," in *2012 IEEE International Symposium on Information Theory Proceedings*. IEEE, july 2012, pp. 1321–1325. [paper] [arXiv]

[C133] J. Villard, P. Piantanida, and S. Shamai, "Secure transmission of a Gaussian source over Gaussian channels with side information," in *2012 IEEE International Symposium on Information Theory Proceedings*. IEEE, july 2012, pp. 1177–1181. [paper]

[C134] A. Zaidi, P. Piantanida, and S. Shamai, "Wyner-Ziv type versus noisy network coding for a state-dependent MAC," in *2012 IEEE International Symposium on Information Theory Proceedings*. IEEE, july 2012, pp. 1682–1686. [paper] [arXiv]

[C135] C. G. Galarza, P. Piantanida, and M. Kountouris, "On the Block Error Probability of Finite-Length Codes in Decentralized Wireless Networks," in *2011 49th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, 28 – 30 September 2011, pp. 1195–1201. [paper]

[C136] A. Behboodi and P. Piantanida, "On the Asymptotic Error Probability of Composite Relay Channels," in *2011 IEEE International Symposium on Information Theory Proceedings*. IEEE, 2011, pp. 1524–1528. [paper]

[C137] M. Ivashkina, I. Andriyanova, P. Piantanida, and C. Poulliat, "Block-Markov LDPC Scheme for Half- and Full-Duplex Erasure Relay Channel," in *2011 IEEE International Symposium on Information Theory Proceedings*. IEEE, 2011, pp. 1469–1473. [paper]

[C138] J. Villard, P. Piantanida, and S. Shamai (Shitz), "Secure Lossy Source-Channel Wiretapping with Side Information at the Receiving Terminals," in *2011 IEEE International Symposium on Information Theory Proceedings*. IEEE, 2011, pp. 1141–1145. [paper] [arXiv]

[C139] A. Zaidi, P. Piantanida, and S. Shamai (Shitz), "Multiple Access Channel with States Known Noncausally at One Encoder and Only Strictly Causally at the Other Encoder," in *2011 IEEE International Symposium on Information Theory Proceedings*. IEEE, 2011, pp. 2801–2805. [paper] [arXiv]

[C140] A. Altieri, L. Rey Vega, C. G. Galarza, and P. Piantanida, "Cooperative Strategies for Interference-Limited Wireless Networks," in *2011 IEEE International Symposium on Information Theory Proceedings*. IEEE, 2011, pp. 1623–1627. [paper] [arXiv]

[C141] S. Yang, P. Piantanida, M. Kobayashi, and S. Shamai (Shitz), "On the Secrecy Degrees of Freedom of Multi-Antenna Wiretap Channels with Delayed CSIT," in *2011 IEEE International Symposium on Information Theory Proceedings*. IEEE, 2011, pp. 2866–2870. [paper] [arXiv]

[C142] M. Kobayashi, P. Piantanida, S. Yang, and S. Shamai (Shitz), "On the Multi-Antenna Wiretap Channel with Delayed CSI at the Transmitter," in *VALUETOOLS – Workshop on Secure Wireless Networks (SECURENETS)*, May 20 2011

[C143] A. Zaidi, P. Piantanida, and S. Shamai (Shitz), "Capacity of Multiple Access Channel with Noncausal and Strictly Causal States at the Encoders," in *NEWCOM++ / COST 2100 Joint Workshop on Wireless Communications (JNCW 2011)*, 2011

[C144] M. Kobayashi, P. Piantanida, S. Yang, and S. Shamai (Shitz), "On the Secrecy Degrees of Freedom of MISO Wiretap Channels with Delayed CSIT," in *NEWCOM++ / COST 2100 Joint Workshop on Wireless Communications (JNCW 2011)*, 2011

[C145] A. Behboodi and P. Piantanida, "Broadcasting over the relay channel with oblivious cooperative strategy," in *2010 48th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, 29 2010-oct. 1 2010, pp. 1098–1103. [paper] [arXiv]

[C146] J. Villard and P. Piantanida, "Secure lossy source coding with side information at the decoders," in *2010 48th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, 29 2010-oct. 1 2010, pp. 733–739. [paper] [arXiv]

[C147] A. Behboodi and P. Piantanida, "On the simultaneous relay channel with collocated relay and destination nodes," in *Modeling and Optimization in Mobile, Ad Hoc and Wireless Networks (WiOpt), 2010 Proceedings of the 8th International Symposium on*, 31 2010-june 4 2010, pp. 437–442

[C148] P. Piantanida and S. Shamai (Shitz), "On the capacity of compound state-dependent channels with states known at the transmitter," in *2010 IEEE International Symposium on Information Theory*. IEEE, june 2010, pp. 624–628. [paper] [arXiv]

[C149] M. Kobayashi, P. Piantanida, S. Yang, and S. Shamai (Shitz), "On the secrecy degress of freedom of the multi-antenna block fading wiretap channels," in *2010 IEEE International Symposium on Information Theory*. IEEE, june 2010, pp. 2563–2567. [paper] [arXiv]

[C150] A. Behboodi and P. Piantanida, "Capacity of a class of broadcast relay channels," in *2010 IEEE International Symposium on Information Theory*. IEEE, june 2010, pp. 590–594. [paper] [arXiv]

[C151] A. Zaidi, S. Shamai (Shitz), P. Piantanida, and L. Vandendorpe, "Bounds on the capacity of the relay channel with noncausal state information at source," in *2010 IEEE International Symposium on Information Theory*. IEEE, june 2010, pp. 639–643. [paper] [arXiv]

[C152] M. Kobayashi, P. Piantanida, S. Yang, and S. Shamai (Shitz), "On the Secrecy Degrees of Freedom of the Multi-Antenna Block Fading Wiretap Channels," in *Future Network and MobileSummit 2010 Conference and Exhibition*, 16-18 June 2010, Florence, Italy 2010

[C153] A. Zaidi, S. Shamai (Shitz), P. Piantanida, and P. Vandendorpe, "Bounds on the Capacity of the Relay Channel with Noncausal State Information at Source," in *Future Network and MobileSummit 2010 Conference and Exhibition*, 16-18 June 2010, Florence, Italy 2010

[C154] J. Villard, P. Bianchi, E. Moulines, and P. Piantanida, "High-rate quantization for the Neyman-Pearson detection of Hidden Markov Processes," in *IEEE Information Theory Workshop 2010 (ITW 2010)*. IEEE, jan. 2010, pp. 1–5. [paper] [arXiv]

[C155] P. Piantanida and S. Shamai (Shitz), "Capacity of compound state-dependent channels with states known at the transmitter," in *2009 IEEE International Symposium on Information Theory*. IEEE, 28 2009-july 3 2009, pp. 1968–1972. [paper]

[C156] A. Behboodi and P. Piantanida, "On the simultaneous relay channel with informed receivers," in *2009 IEEE International Symposium on Information Theory*. IEEE, 28 2009-july 3 2009, pp. 1179–1183. [paper]

[C157] P. Piantanida and S. Shamai (Shitz), "Capacity of compound state-dependant channels with state information at the transmitter," in *Proc. of NEWCOM - ACoRN Joint Workshop*, March 30-April 1,Barcelona, Spain 2009

[C158] P. Layec, P. Piantanida, R. Visoz, and A. Berthet, "Transceiver design for sum-MSE optimization in MIMO-MAC with imperfect channel estimation," in *Signals, Systems and Computers, 2008 42nd Asilomar Conference on*, oct. 2008, pp. 321–325. [paper]

[C159] A. Venkiah, P. Piantanida, C. Poullia, P. Duhamel, and D. Declercq, "Rateless coding for quasi-static fading channels using channel estimation accuracy," in *2008 IEEE International Symposium on Information Theory*. IEEE, july 2008, pp. 2257–2261. [paper]

[C160] P. Layec, P. Piantanida, R. Visoz, and A. Berthet, "Capacity of channels with multistage successive refinement of quantized feedback information via noisy links," in *2008 IEEE 9th Workshop on Signal Processing Advances in Wireless Communications*. IEEE, july 2008, pp. 176–180. [paper]

[C161] ——, "Capacity bounds for MIMO multiple access channel with imperfect channel state information," in *2008 IEEE Information Theory Workshop*. IEEE, May 2008, pp. 21–25. [paper]

[C162] P. Piantanida, S. Sadough, and P. Duhamel, "On the Outage Capacity of a Practical Decoder Using Channel Estimation Accuracy," in *2007 IEEE International Symposium on Information Theory*. IEEE, june 2007, pp. 2281–2285. [paper] [arXiv]

[C163] S. Sadough, P. Piantanida, and P. Duhamel, "MIMO-OFDM optimal decoding and achievable information rates under imperfect channel estimation," in *2007 IEEE 8th Workshop on Signal Processing Advances in Wireless Communications*. IEEE, june 2007, pp. 1–5. [paper] [arXiv]

[C164] P. Piantanida and P. Duhamel, "Dirty-Paper Coding without Channel Information at the Transmitter and Imperfect Estimation at the Receiver," in *2007 IEEE International Conference on Communications*. IEEE, june 2007, pp. 5406–5411. [paper] [arXiv]

[C165] ——, "On the Capacity of the Fading MIMO Broadcast Channel Without Channel Information at the Transmitter and Imperfect Estimation at the Receivers," in *2007 IEEE International Conference on Acoustics, Speech and Signal Processing - ICASSP '07*, vol. 3. IEEE, april 2007, pp. III–409–III–412. [paper]

[C166] ——, "Achievable Rates for the Fading MIMO Broadcast Channel with Imperfect Channel Estimation," in *Communication, Control, and Computing (Allerton), 2010 44th Annual Allerton Conference on*, Sep. 27-29 2006

[C167] S. Sadough, P. Piantanida, and P. Duhamel, "Achievable Outage Rates with Improved Decoding of BICM Multiband OFDM under Channel Estimation Errors," in *2006 Fortieth Asilomar Conference on Signals, Systems and Computers*. IEEE, 29 2006-nov. 1 2006, pp. 1806–1810. [paper] [arXiv]

[C168] P. Piantanida, G. Matz, and P. Duhamel, "Outage Behavior of Discrete Memoryless Channels Under Channel Estimation Errors," in *Proceedings of the International Symposium on Information Theory and its Applications (ISITA)*, 2006, pp. 417–422. [paper] [arXiv]

[C169] ——, "Estimation-Induced Outage Capacity of Ricean Channels," in *2006 IEEE 7th Workshop on Signal Processing Advances in Wireless Communications*. IEEE, july 2006, pp. 1–5. [paper]

[C170] A. Zaidi and P. Piantanida, "MAC Aware Coding Strategy for Multiple User Information Embedding," in *2006 IEEE International Conference on Acoustics Speed and Signal Processing Proceedings*, vol. 5. IEEE, may 2006, pp. V–393–V–396. [paper]

[C171] A. Zaidi, P. Piantanida, and P. Duhamel, "Scalar scheme for Multiple User Information Embedding," in *Proceedings. (ICASSP '05). IEEE International Conference on Acoustics, Speech, and Signal Processing, 2005.*, vol. 2. IEEE, 18-23, 2005, pp. 5–8. [paper]

[C172] P. Piantanida, S. Lasaulce, and P. Duhamel, "Broadcast Channels with Noncausal Side Information: Coding Theorem and Application Example," in *Proc. Winterschool on Coding and Information Theory*, Feb. 20-25, , Bratislava, Slovakia 2005

[C173] P. Piantanida and C. Estienne, "Maximum Entropy Good-Turing Estimator for Language Modeling," in *8th European Conference on Speech Communication and Technology (Eurospeech 2003)*. ISCA, September 1-4, Geneva, Switzerland 2003, pp. 2277–2280. [paper]

Invited Papers in proceedings of international conferences

[InvC1] M. Benammar and P. Piantanida, "On Robust Deep Neural Decoders," in *2019 53rd Asilomar Conference on Signals, Systems, and Computers*. IEEE, Oct 2019, pp. 527–531. [paper]

[InvC2] K. Wan, D. Tuninetti, P. Piantanida, and M. Ji, "On Combination Networks with Cache-aided Relays and Users," in *WSA 2018; 22nd International ITG Workshop on Smart Antennas*, March 2018, pp. 1–7. [paper] [arXiv]

[InvC3] M. Benammar and P. Piantanida, "Optimal Training Channel Statistics for Neural-based Decoders," in *2018 52nd Asilomar Conference on Signals, Systems, and Computers*. IEEE, Oct 2018, pp. 2157–2161. [paper]

[InvC4] K. Wan, M. Ji, P. Piantanida, and D. Tuninetti, "Recent Advances on Combination Networks with End-User Caches," in *Information Theory and Applications Workshop (ITA), Catamaran Resort, Pacific Beach, San Diego.*, Feubrary 11th – 16th 2018

[InvC5] K. Wan, D. Tuninetti, M. Ji, and P. Piantanida, "State-of-the-art in cache-aided combination networks," in *2017 51st Asilomar Conference on Signals, Systems, and Computers*. IEEE, Oct 2017, pp. 641–645. [paper]

[InvC6] K. Wan, D. Tuninetti, P. Piantanida, and M. Ji, "On Cache-aided Relay Networks," in *The 22nd International ITG Workshop on Smart Antennas (WSA 2018), Bochum, Germany*, March 14 - 16 2018

[InvC7] K. Wan, D. Tuninetti, and P. Piantanida, "A novel index coding scheme and its application to coded caching," in *2017 Information Theory and Applications Workshop (ITA)*. IEEE, Feb 2017, pp. 1–6. [paper] [arXiv]

[InvC8] K. Wan, D. Tuninetti, M. Ji, and P. Piantanida, "State-of-the-art in cache-aided combination networks," in *2017 51st Asilomar Conference on Signals, Systems, and Computers*. IEEE, Oct 2017, pp. 641–645. [paper]

[InvC9] A. Bunin, Z. Goldfeld, H. H. Permuter, S. Shamai (Shitz), P. Cuff, and P. Piantanida, "Semantically-Secured Message-Key Trade-Off over Wiretap Channels with Random Parameters," in *Proceedings of the 2nd Workshop on Communication Security*, M. Baldi, E. A. Quaglia, and S. Tomasin, Eds. Cham: Springer International Publishing, 2018, pp. 33–48. [paper]

[InvC10] G. Bassi, P. Piantanida, and S. Shamai (Shitz), "The Role of Noisy Feedback in Secure Communications," in *EuCNC'2015 - European Conference on Networks and Communications*, 2015

[InvC11] A. Altieri, P. Piantanida, L. Vega, and C. Galarza, "A stochastic geometry approach to distributed caching in large wireless networks," in *2014 11th International Symposium on Wireless Communications Systems (ISWCS)*. IEEE, Aug 2014, pp. 863–867. [paper]

[InvC12] C. He, S. Yang, and P. Piantanida, "On the capacity of the fading broadcast channel with state feedback," in *2014 6th International Symposium on Communications, Control and Signal Processing (ISCCSP)*. IEEE, May 2014, pp. 182–185. [paper]

[InvC13] M. Benammar, P. Piantanida, and S. Shamai (Shitz), "Dirty-paper Coding Techniques for Compound MISO Broadcast Channels: A DoF Analysis," in *Proceedings of the 9th International Conference on Cognitive Radio Oriented Wireless Networks*. ICST, June 2014. [paper]

[InvC14] A. Adhikary, P. Piantanida, M. Kobayashi, H. Papadopoulos, and G. Caire, "CSI Dissemination for MU-MIMO Schemes Based on Outdated CSI," in *2012 50th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, 2012, pp. 1672–1679. [paper]

[InvC15] A. Behboodi and P. Piantanida, "Selective Coding Strategy for composite relay channels," in *2012 5th International Symposium on Communications, Control and Signal Processing*. IEEE, may 2012, pp. 1–6. [paper]

[InvC16] ——, "On the asymptotic spectrum of the error probability of composite networks," in *2012 IEEE Information Theory Workshop*. IEEE, sept. 2012, pp. 148–152. [paper]

[InvC17] A. Altieri, L. Rey Vega, C. Galarza, and P. Piantanida, "On the balance between cooperation and interference in dense wireless networks," in *Wireless Communication Systems (ISWCS), 2012 International Symposium on*, aug. 2012, pp. 1009–1013. [paper]

[InvC18] J. Villard, P. Piantanida, and S. Shamai (Shitz), "Hybrid Digital/Analog Schemes for Secure Transmission with Side Information," in *2011 IEEE Information Theory Workshop*. IEEE, 16 – 20 October 2011, pp. 678–682. [paper] [arXiv]

[InvC19] J. Villard and P. Piantanida, "Secure Distributed Lossless Compression with Side Information at the Eavesdropper," in *VALUETOOLS – Workshop on Secure Wireless Networks (SECURENETS)*, May 20 2011. [paper]

[InvC20] A. Zaidi, S. Shamai (Shitz), P. Piantanida, and L. Vandendorpe, "On the capacity of a class of relay channels with orthogonal components and noncausal state information at source," in *IEEE 5th International Symposium on Wireless Pervasive Computing 2010*. IEEE, May 2010, pp. 528–533. [paper]

Proceedings of national conferences

[NatC1] R. Mouravieff, T. Herserant, A. Satouf, H. A. Boubacar, P. Piantanida, V. Guigue, S. Lamprier, and B. Piwowarski, "DQE : Génération de données diversifiées par agents pour le question-réponse sur tables," in *Actes de la 21e Conférence en Recherche d'Information et Applications. Volume 1 : articles scientifiques originaux*, N. HERNANDEZ and S. QUINIOU, Eds. Nantes, France: ATALA, Jun. 2026, pp. 94–106. [paper]

[NatC2] F. Alberge, C. Feutry, P. Piantanida, and P. Duhamel, "A Simple Statistical Method to Detect Covariate Shift," in *Colloque GRETSI - Traitement du Signal et des Images (GRETSI 2019)*, August 2019

[NatC3] M. Picot, F. Messina, P. Piantanida, and F. Labeau, "Modelling the Uncertainty of a Deep Neural Network Enhances its Adversarial Robustness," in *Colloque GRETSI - Traitement du Signal et des Images (GRETSI 2019)*, August 2019

[NatC4] V. Matias, L. Rey Vega, and P. Piantanida, "Algoritmo Iterativo Multitarea con un Enfoque de Teoría de Información," in *Proc. of XVII Workshop in Information Processing and Control (RPIC 2017)*, 20 – 22 September 2017

[NatC5] A. Altieri, L. Rey Vega, P. Piantanida, and C. G. Galarza, "Performance of Cooperative Strategies on Dense Wireless Networks," in *Proc. of XIV Workshop in Information Processing and Control (RPIC 2011)*, 16 – 18 November 2011

[NatC6] J. Villard and P. Piantanida, "Codage de source sous contrainte de sécurité avec information adjacente aux récepteurs," in *XXIIIe Colloque GRETSI - Traitement du Signal et des Images (GRETSI 2011)*, 5-8 September 2011

[NatC7] A. Behboodi and P. Piantanida, "Cooperative Strategies for Simultaneous Relay Channels," in *XXIIIe Colloque GRETSI - Traitement du Signal et des Images (GRETSI 2011)*, 5-8 September 2011

[NatC8] P. Piantanida and C. Estienne, "Improving Backing-off by Considering Extended Contexts," in *Proc. of Workshop in Information Processing and Control (RPIC)*, October 10-14, San Nicolas, Argentina 2003

[NatC9] P. Piantanida, N. Barraza, and C. Estienne, "Una nueva familia de estimadores de Probabilidades de n-gramas en Modelos Coersitivos," in *Proc. of Workshop in Information Processing and Control (RPIC)*, October 10-14, San Nicolas, Argentina 2003

[NatC10] C. Estienne, P. Pelle, and P. Piantanida, "A Front-end for Speech Recognition Systems Using Phase-Locked Loops," in *Proc. of Workshop in Information Processing and Control (RPIC)*, October 10-14, San Nicolas, Argentina 2001, pp. 88–91

Book chapters and technical reports

[B1] P. Piantanida and L. R. Vega, *Information Bottleneck and Representation Learning*. Cambridge University Press, 2021, pp. 330–358, doi:10.1017/9781108616799.012. [paper]

[B2] F. Regol, J. Chataoui, B. Charpentier, M. Coates, P. Piantanida, and S. Gunnemann, "Predicting Probabilities of Error to Combine Quantization and Early Exiting: QuEE," 2024. [paper] [arXiv]

[B3] G. Pichler, P. Piantanida, and G. Koliander, "On the estimation of information measures of continuous distributions," *CoRR*, vol. abs/2002.02851, 2020. [paper] [arXiv]

HDR, Ph.D. and M.Sc Thesis

[R1] P. Piantanida, "Information-theoretic methods: From communication to decision, privacy and learning, at the heart of data sciences," HDR (Habilitation à diriger des recherches), Université Paris-Saclay, Gif-sur-Yvette, France, Tech. Rep., May 2019. [paper]

[R2] ——, "Multi-user information theory: State information and imperfect channel knowledge," Ph.D. dissertation, Université Paris-Sud, Gif-sur-Yvette, France, May 2007. [paper]

[R3] ——, "Statistical language modeling using maximal entropy methods," Master's thesis, Buenos Aires University (UBA), Buenos Aires, Argentina, September 2003